



Prediction of cement-based mortars compressive strength using machine learning techniques

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Abstract

The application of artificial neural networks in mapping the mechanical characteristics of the cement-based materials is underlined in previous investigations. However, this machine learning technique includes several major deficiencies highlighted in the literature, such as the overfitting problem and the inability to explain the decisions. Hence, the present study investigates the applicability of other common machine learning techniques, i.e., support vector machine, random forest (RF), decision tree, AdaBoost and k-nearest neighbors in mapping the behavior of the compressive strength (CS) of cement-based mortars. To this end, a big experimental database has been compiled based on experimental data available in the literature considering, namely the cement grade, which is an important parameter for the modeling of mortar's CS. Other important parameters are namely the age, the water-to-binder ratio, the particle size distribution of the sand and the amount of plasticizer. Many models based on the influential factors affecting machine learning techniques have been developed, and their prediction capacities have been assessed using performance indexes. The present research highlights the potential of AdaBoost and RF models as useful tools which can assist in mortar design and/or optimization. In addition, mapping the development of mortar characteristics can assist in revealing the influence of the different mortar mix parameters on the compressive strength.

Keywords Predictive models · Cement · Compressive strength · Machine learning · Metakaolin · Mortar

Abbreviations

AI	Artificial intelligence
ANN	Artificial neural network
SA	Sensitivity analysis
B/S	Binder-to-sand ratio
DT	Decision tree
RF	Random forest
CG	Cement grade
CS	Compressive strength
kNN	K-nearest neighbors

MAPE	Mean absolute percentage error
MK/B	Metakaolin percentage in relation to total binder
R^2	Coefficient of determination
RMSE	Root-mean-square error
SP	Superplasticizer
SVM	Support vector machine
CAM	Cosine amplitude method
W/B	Water-to-binder ratio
PL	Polynomial kernel
LN	Linear kernel
OOB	Out-of-bag
RBF	Radial basis function kernel
SIG	Sigmoid kernel
C	Regularization parameter
γ	Kernel width
AS	Age of specimen
d	Polynomial kernel degree
STD	Standard deviation
R_{ij}	Strength of the relation
SC	Soft computing

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1 Introduction

During the last decades, there has been great need and interest in research, to produce eco-friendly building materials, in order to support the high demand by public and private structures. Concrete remains the most commonly used man-made material on earth. Based on the fact that the production of cement has substantial economic and environmental impact, research has focused on both the reduction of the environmental footprint and the cost reduction of cement-based materials. An important scientific field of research in the production of mortar materials is the partial replacement of Portland cement with Metakaolin (MK).

MK is cement's surrogate material, produced during the calcination of kaolin clay such as silica and alumina at high temperatures (from 700 to 900 °C), and it has been used to replace a portion of cement's percentage in mortars, where the percentage of replacement in relation to the total binder varies from 10% to more than 50% [1–4]. The use of MK as a replacing material of Portland cement in mortars has resulted in the enhancement of their mechanical and durability performance [5–7]. Specifically, there is a noticeable increase in the mortar's compressive strength (CS) mainly during the early curing days at normal temperatures due to the pozzolanic reaction, the fineness of MK's aggregates and the acceleration of cement hydration [8]. Furthermore, towards the direction of a holistic environmental approach, in addition to this new trend of replacing the cement with metakaolin in mortar, technology aims to reduce the large amount of carbon dioxide (CO₂) emissions during cement production.

The inclusion of MK in mortar mix renders the estimation of mortar's CS, via an analytical formula, more difficult, than in the case of a conventional mortar, where the mix parameters are less, than those in cement–metakaolin mortars. This is mainly due to the large number of mortar components, as well as due to the strongly nonlinear behavior exhibited by the CS in relation to the mix parameters. To this end, in cases where the input and output values cannot be directly associated by simple equations, soft computing (SC) techniques, often, provide valuable alternatives, in engineering problem solving context [9–27]. Such techniques, due to the nonlinear and complex relationships between independent and dependent variables, can be performed in the field of computational structural engineering and for materials with a sufficiently high degree of accuracy, as long as a representative database of experimental results is available. This way, many new intelligent techniques can be applied and introduced for different applications of engineering.

So far, available literature includes mainly publications concerning the application of the SC techniques such as

artificial neural networks (ANNs) in materials science. Specifically, such computational techniques were used for the prediction of the CS and modulus of elasticity of cement-based materials. Detailed and in-depth state-of-the-art reports on the use of ANNs in the modeling of concrete materials can be found in the literature [28–41]. In the last decade, other than ANN models, a plethora of computational models such as fuzzy logic and genetic algorithms have also been used for the modeling of cement-based materials' mechanical characteristics [30, 37, 38, 42–49].

Since the number of applied SC techniques in the field of cement-based materials' mechanical characteristics is limited to only a few ANN techniques, there is a need to investigate the applicability and suitability of the other SC techniques. Hence, in the work presented herein, the application of the SC techniques in predicting cement-based mortar's CS is investigated. Specifically, five different SC predictive models are presented, namely the support vector machine (SVM), the decision tree (DT), the random forest (RF), the AdaBoost and the k-nearest neighbors (KNN), with their most effective parameters being defined, allowing to efficiently investigate and predict the cement-based mortars' CS. Since the term of model generalization is considered as a very important issue in any prediction/simulation study, the authors used the most comprehensive database of cement-based materials' mechanical characteristics. The reported database comprises 424 data samples with the most effective parameters on the CS values of cement-based materials. To the author's best knowledge the database reported in this paper is the biggest to date reported database which was compiled by extending a previous database reported by the authors [38]. The database includes cement grade as an input parameter, which has a significant effect on the compressive strength of mortar, and which has rarely been considered before as a critical input parameter. The trained and developed model has holistically mapped the compressive strength of mortar. Eventually, the performance capacity of the proposed models is compared with each other in order to choose and introduce the best predictive model for solving mechanical characteristics of cement-based material.

Considering the above, the manuscript is organized into six sections, including this introductory section. In Sect. 2, the significance of research in the area of the cement-based mortars' CS is discussed. Then, Sect. 3 describes the principles of the applied SC techniques, the research process of this study, the behavior of cement-based mortars, the developed database, a sensitivity analysis of the input parameters, and performance indices used to evaluate the proposed models. Section 4 discusses how the SC models are applied and proposed to predict the cement-based mortars' CS. Afterward, Sect. 5 presents the results of the

proposed models in estimating the cement-based mortars' CS and conclusions of the current research, respectively.

2 Research significance

In the last decades, there has been great demand for mass production of eco-friendly, low cost and high-performance building materials, which are also durable under unfavorable environmental conditions. A cement-based material such as mortar which is a flagship building material and whose environmental requirements resulted in the inclusion of more components and additives than previous conventional mortars. However, the large number of the mortar's mixture parameters, as well as their strongly nonlinear relation/contribution to the mortar's mechanical characteristics, such as the CS, renders the analytical formulae for the numerical estimation of the mortar's CS, extremely complicated. To this end, nonconventional methods become an important tool, providing solutions to the above complex optimization problem. In this context, in the work presented here, the most commonly used SC techniques such as SVM, DT, RF, AdaBoost and KNN have been proposed for the estimation of mortar's CS, which is a key parameter for the reliable design of masonry buildings, as well as for their conservation in many countries in which cement-based mortars have been extensively used in the original construction. The optimum predictive model among the proposed SC models has proven to be very successful, exhibiting reliable predictions and, above all, revealing its strongly nonlinear mechanical nature.

3 Materials and methods

3.1 Principles of the used models and research process

3.1.1 Support vector machines

Support vector machine (SVM) has been introduced in recent years as an innovative supervised learning method. It works on the basis of the structural risk minimization principle and statistical learning theory [50]. SVM uses training data in order to implicitly map the original input space into a high-dimensional feature space. As a result, the optimal hyperplane within the feature space is specified through the maximization of the class boundaries [51]. In this system, support vectors are those training points that are positioned closest to the optimal hyperplane. New data are classified when decision surface is achieved.

To have an effective SVM training and to have precision in classification processes, it is important to select the kernel

function properly [52]. In SVM, four types of kernel function groups are typically utilized: polynomial kernel (PL), linear kernel (LN), radial basis function (RBF) kernel, and sigmoid kernel (SIG). Among them, LN is a specific case of RBF, while the behavior of SIG is similar to RBF in certain parameters [53]. Keerthi and Lin [54] maintained that in cases where RBF is employed, LN is no longer needed. In general, no evidence has shown the higher accuracy of SIG in classification, compared with RBF [55]. Based on Zhu et al.'s [56] findings, better interpolation capability is the main advantage of RBF over PL. On the other hand, RBF may be incapable of providing longer-range extrapolation. PL, on the contrary, has shown higher extrapolation capacity at lower-order degrees; however, for an acceptable interpolation, higher order degrees are needed. The SVM model performance is dependent upon the selection of the kernel parameters. In case of the RBF-SVM model, two parameters are required to be specified, namely the regularization parameter (C) and the kernel width (γ); while in case of the PL-SVM model, not only C and γ , but also the polynomial kernel degree (d) must be determined. C is responsible for controlling the trade-off between margin and training errors, thereby controlling the model overfitting. Large values of C result in few training errors, while small values result in a larger margin, which causes an increase in the number of training errors [57]. On the other hand, γ is responsible for controlling the nonlinearity degree of the SVM model, and parameter d determines the polynomial kernel degree. In the context of data mining, a significant topic is the process of selecting the optimal pairs of parameters in a way to obtain the best classification result [58].

3.1.2 Decision tree

A decision tree (DT) refers to a hierarchical model containing decision rules based on which independent variables are split recursively into homogeneous zones [59]. DT is mainly aimed at exploring a set of decision rules applicable to the prediction of an outcome considering a set of inputs. In cases where the target variables are continuous or discrete, DT is termed regression tree or classification tree, respectively [60]. Numerous researchers have used DT effectively in different real conditions for prediction and/or classification purposes [61].

The most important benefit of DT is its high capacity in modeling complicated relationships among existing variables. The DT models have the capacity of incorporating both continuous and categorical variables with no strict assumption, through considering how the data are distributed [62]. Moreover, it is not difficult to build a DT and the resultant models are simply interpretable. In addition, DT is able to provide available perfect information in regard to the relative significance of input parameters [63].

On the other hand, DT suffers from two disadvantages: it is vulnerable to noisy data and it does not allow for multiple output attributes [64].

Numerous algorithms have been introduced by different scholars to construct DT models, including ID3 [65], the classification and regression tree (CART) [66], C4.5 [67] and Chi-square automatic interaction detector decision tree (CHAID) [68]. In a tree structure, a root node exists, as well as leaf nodes and internal nodes. Among them, the former consists of all input data. Each internal node may hold two or more branches, and it is accompanied with a decision function. Each leaf node shows the output of a specific input vector.

DT modeling is conducted in two steps: tree building and tree pruning [69]. The former step starts when identifying the input variable that has the maximum gain ratio as the root node of DT. After that, splitting of the training dataset is carried out by the basis of the root values, and then subnodes are formed. In cases where input variables are discrete, for each possible value, a subnode of the tree is formed. Whereas in some cases of continuous input variables, two subnodes are formed on the basis of a threshold defined through the threshold-finding process [67]. Afterward, in the case of all subnodes, the gain ratio is computed, and the procedure is repeated successively until all examples within a given node belong to a single class. Such nodes are then termed ‘leaf nodes,’ which are labeled as class values.

A tree achieved at the building step may hold many branches, leading to an overfitting problem [70]; as a result, there is a need to prune the tree in order to achieve higher classification accuracy for new data. Trees can be pruned in two ways: pre-pruning and post-pruning. In the former type, once a definite criterion is met, the tree growth is stopped; while in the latter one, first, the full tree is formed, and the ending subtrees are then replaced by leaves on the basis of the error comparison of the tree prior to and after the subtrees replacement.

3.1.3 AdaBoost

AdaBoost [71] is known as an efficient ensemble learning algorithm iteratively choosing a number of classifier instances by performing weight distribution, adaptively, over training instances. This algorithm linearly combines the selected classifier examples with the aim of forming an overall ensemble. Even in cases where many base classifier instances are implemented in the model, the ensembles formed based on AdaBoost hardly show an overfitting problem [72]. Such ensembles minimize an exponential loss function through fitting a stage-wise additive model. When the classification error is minimized, it is implied that a nondifferentiable, nonsmooth cost function has been

optimized; it could be approximated in the most proper way by means of an exponential loss [72]. Consequently, it can be said that AdaBoost works highly efficiently when addressing many types of classification problems.

AdaBoost makes alterations to the original training set distribution. Let ϵ_t be the misclassification rate at trial t ; in this condition, the weights of those misclassified samples that exist within the training set are altered by factor $\beta_t = (1 - \epsilon_t)/\epsilon_t$. The total sum of the adjusted weights is normalized to 1. A combination of the classifiers $C_1, \dots, C_t$ is created based on the weighted voting, where C_t is weighted by $\log(\beta_t)$. The trials end if $\epsilon_t \leq 0.5$ (i.e., the precision of the classifier is lower than a random guess), and then trial T is changed to $t - 1$. On the other hand, if $\epsilon_t = 0$ (which shows 100% accuracy), trial T becomes t . In such process, AdaBoost forces the ensemble classifiers (which are newly created) to be entirely focused upon ‘harder’ cases. By taking into consideration the precision of each classifier, the voting for the final labels is weighted. AdaBoost in the present paper is executed with the use of DT in which the trees are built using the information gain ratio [67]. A total view of AdaBoost algorithm is presented in Fig. 1.

3.1.4 Random forest

In random forest (RF), a group of classifiers is constructed through changing the training set with the use of the bagging technique [73]. This technique forms fresh training sets through resampling from the original data set for n times (where n stands for the number of samples that exist within the original training set) randomly with replacement. In other words, the newly selected sample is not eliminated from the data set in the following draw. As a result, a number of the training samples are selected twice or more, whereas some others will not be selected at all in a new set. Bagging improves the precision of classification through reducing the variance of the classification errors; that is, bagging taps on a classifier instability. Here, ‘instability’ means that a slight change to the training samples leads to considerable changes in the level of precision. A majority vote is utilized for the aim of combining the classifiers, and each classifier’s vote holds the same weight. For a tie, the decision making is done by prescribed rules or in a random way. RF makes use of the impurity Gini index and forms multiple trees [74]. Although, at the time of forming a tree, RF attempts to find only a random sub set of the input features (bands) at each splitting node, and the tree can develop completely with no need for pruning. Remember that RF is computationally light because merely a part of the input features is employed and there is no need for pruning process. The computing time is in the order of $T\sqrt{M} \log(N)$ where T stands for the number of trees, N

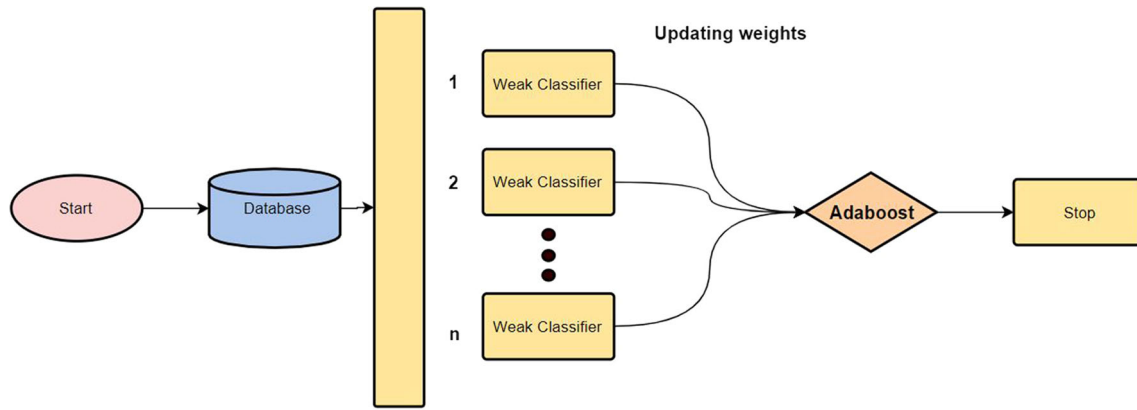


Fig. 1 AdaBoost flowchart

denotes the number of training samples, and M is the number of bands that are utilized in each split [75]. Moreover, if a separate test set cannot be found, an out-of-bag method is applicable. For each newly produced training set, 33% of the samples are left out in a random way, which are known as the out-of-bag (OOB) samples. The rest of them, i.e., in-the-bag samples, are applied to tree construction processes. To predict the accuracy level, for each sample, the votes are counted every time that the sample belongs to OOB samples. The final label is determined based on the majority vote. Only 33% of the constructed trees are allowed to vote for each case. These OOB error predictions are unbiased in numerous tests [75]. User typically defines the number of features for each split, which is not sensitive to the algorithm. The majority vote is applied to combining the ensemble classifiers decisions. A detailed explanation about the RF can be found in the study of Breiman [75]. Gislason et al. [76] also provides a summary in this regard. Figure 2 shows a flowchart of the RF process.

3.1.5 The k-nearest neighbors

The k-nearest neighbors (kNN) in machine learning is recognized widely as a lazy algorithm [77, 78]. In kNN, all observations that the algorithm reads are firstly stored, after that, other observations are predicted on the basis of distance functions. kNN mainly aims at calculating a numerical target that is averaged from the k 's nearest neighbors [79]. Furthermore, to calculate such distance, the inverse distance weighted average is employed [80]. In regression-related problems, three distance functions are utilized by kNN for the purpose of calculating the distance between the neighbors, as expressed in the following:

$$\text{Euclidean function : } \sqrt{\sum_{i=1}^f (x_i - y_i)^2} \quad (1)$$

$$\text{Manhattan function : } \sum_{i=1}^f |x_i - y_i| \quad (2)$$

$$\text{Minkowski function : } \left(\sum_{i=1}^f (|x_i - y_i|)^q \right)^{1/q} \quad (3)$$

where x_i and y_i stand for the i th dimensions of the x and y points, respectively, q represents the order between x and y points.

3.1.6 Research process

The present study collected 424 laboratory data from the available literature (referenced below) to develop intelligent models for the prediction of cement-based mortars compressive strength (CS). After reviewing the data, the importance of each parameter was examined and the relationship between them was identified. The data were randomly divided into two categories i.e., training and testing. The training data (80% of the total data) were used to train the intelligent models, and then the performance of the proposed models was assessed using the testing data (the remaining 20%). Five different machine learning models were developed in this study to predict cement-based mortars CS. Each artificial intelligence (AI) model has special features, which are best achieved using parametric analysis. Finally, a comparison was made between the proposed models and the model is demonstrated as a powerful, applicable and simple AI technique for the prediction of cement-based mortars CS. Figure 3 presents the general process of the current research.

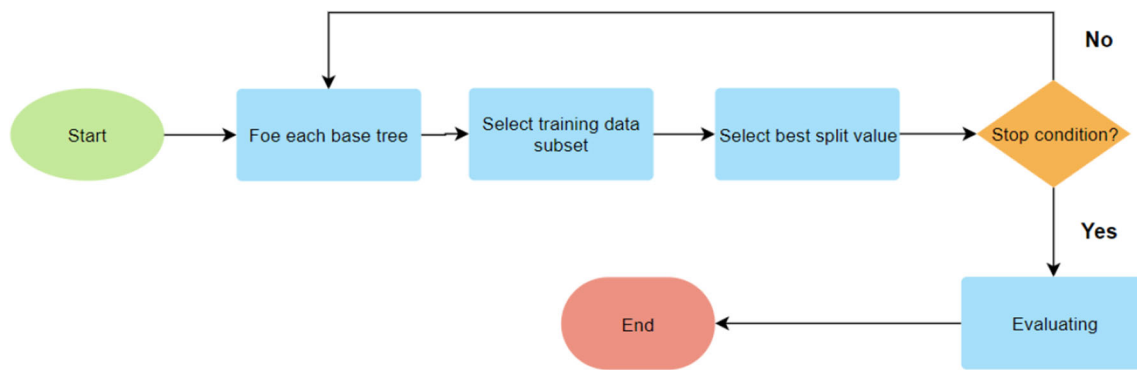


Fig. 2 RF process

3.2 Cement-based mortars database

It is common practice for researchers, during the developing process of a forecast model, to pay particular attention to the computational model itself, at the expense of the actual quality of the database used for the development, training and validation of the model. Although research related to new computational models is undeniably of high importance and adds significant value in advancing in the field, the authors believe that, since the

ultimate goal is a reliable forecast, the reliability of the database should be of utmost importance and should be thoroughly investigated in this regard. In fact, a reliable database should not only comprise reliable data, but also a sufficient amount of data covering the full range of parameter values. Aiming to avoid any misinterpretation of the term ‘sufficient,’ it is clarified that ‘sufficient’ amount of data is not only large amount of data, but, rather, data that covers a wide range of combinations of input parameters, thus assisting in the model’s ability to simulate the problem. The demand for reliable and capable database is crucial in the case of databases compiled using experimental results. In this case, high variance between experimental values has been frequently noticed, not only between testing campaigns conducted by different research teams and laboratories, but even between datasets that derive from experiments conducted on specimens of the same synthesis, produced by the same technicians, cured under the same conditions, adopting the same testing standards and using the same testing instruments.

Thus, a reliable database, which also covers the full range of parameters and combinations of parameters, in addition to contributing to the development of reliable mathematical models, will also decisively contribute to the reliable comparison between different forecast computational models.

Considering the above discussion, a big and reliable experimental database has been compiled based on data available in literature. Specifically, for the development and training of the proposed SC models presented here, an experimental database has been compiled which consists of 424 datasets (Table 1) based on experimental results of 20 published research works on the behavior of cement-based mortars, with and without metakaolin, as well as superplasticizers. These 20 research works have been conducted based on the same standards and cover the full range of parameters affecting the CS of mortars (note in particular the usage of high-strength cement and the wide range of CS values, from 4 to 115 MPa, which are of interest for

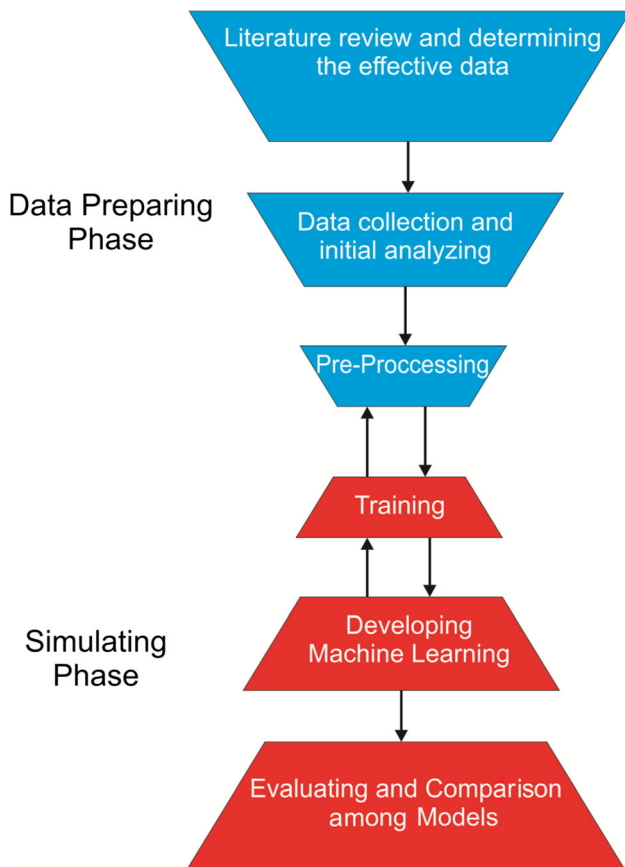


Fig. 3 The general process of current research

Table 1 Data from experiments published in literature

No	Reference	No. of samples	Parameters		CS (MPa)
			MK	SP	
1	Vu et al. [1]	144	✓	✓	15.45–54.90
2	Courard et al. [81]	15	✓		27.40–71.20
3	Sumasree and Sajja [82]	35	✓		24.93–35.71
4	Batis et al. [2]	16	✓		17.60–69.70
5	Kadri et al. [6]	12	✓	✓	35.21–99.17
6	Mardani-Aghabaglou et al. [83]	6	✓		37.27–56.76
7	Potgieter-Vermaak and Potgieter [84]	20	✓		23.40–92.80
8	Cizer et al. [85]	5			47.94–66.23
9	Al-Chaar et al. [86]	3			6.05–10.73
10	Khatib and Wild [8]	4	✓		40.08–56.76
11	Curcio et al. [87]	15	✓	✓	85.99–115.25
12	Cyr et al. [88]	8	✓		18.10–67.50
13	Saidat et al. [89]	40	✓		4.10–64.50
14	Ashok et al. [90]	4			26.37–48.08
15	Geng and Li [91]	20	✓		6.01–78.54
16	Pavlikova et al. [5]	27	✓	✓	44.98–83.16
17	Khater [92]	6	✓		9.75–11.78
18	Parande et al. [93]	20	✓		22.00–67.00
19	Mansour et al. [94]	12	✓	✓	22.99–92.50
20	Lee et al. [95]	12	✓	✓	33.08–63.00
	Total	424			4.10–115.25

different applications). It should be noted that an important parameter used for the modeling of mortar's CS, is the cement grade. Despite the plethora of published works on the computational modeling of mortar's mechanical characteristics such as the CS, the cement grade is not included as an input parameter in any published models, although its impact is considerable. Other important parameters are namely the age and the water-to-binder ratio (linked to the particle size distribution of the sand and the presence of a plasticizer). Detailed and in-depth description of the database can be found in the next section where a short review on the behavior of cement-based mortars with metakaolin is presented. Detailed and in-depth the database is presented in Table 6 of Appendix.

Based on the aforementioned database for the training of the developed computational predictive models, there are six (6) input parameters, namely the age of specimen (AS), the cement grade (CG), the metakaolin percentage in relation to total binder (MK/B), the water-to-binder ratio (W/B), the superplasticizer (SP), which is the percentage of the addition of superplasticizer in relation to the total binder (%w/w), and the binder-to-sand ratio (B/S). The output training parameter is the value of the CS of the mortar specimens.

The mean values of the above input and output parameters of the database, together with their minimum,

maximum values, as well as standard deviation (STD) values, are presented in Table 2. At this point, it should be mentioned that some of the variables may be dependent on each other. Subsequently, the correlation coefficients between all variables have been specified and are presented in Table 3, as well as in Fig. 4. High negative or positive values of the correlation coefficient between the input variables may result in poor efficiency of the methods and in a difficulty in construing the effects of the variables on the response. It is shown that there are not many significant correlations between the independent input variables. Based on the correlation matrix values, it is evident that there is strong relation between the value of mortar CS and the value of the input parameter of the W/B as well as with the value of the CG. The parameter SP is also quite relevant but some degree of correlation with W/B is found, as expected. Finally, there is some correlation between age and CS, also as expected (this correlation is not strong, because different mixes and ages are considered. Obviously, that the correlation would be strong for a specific mix composition).

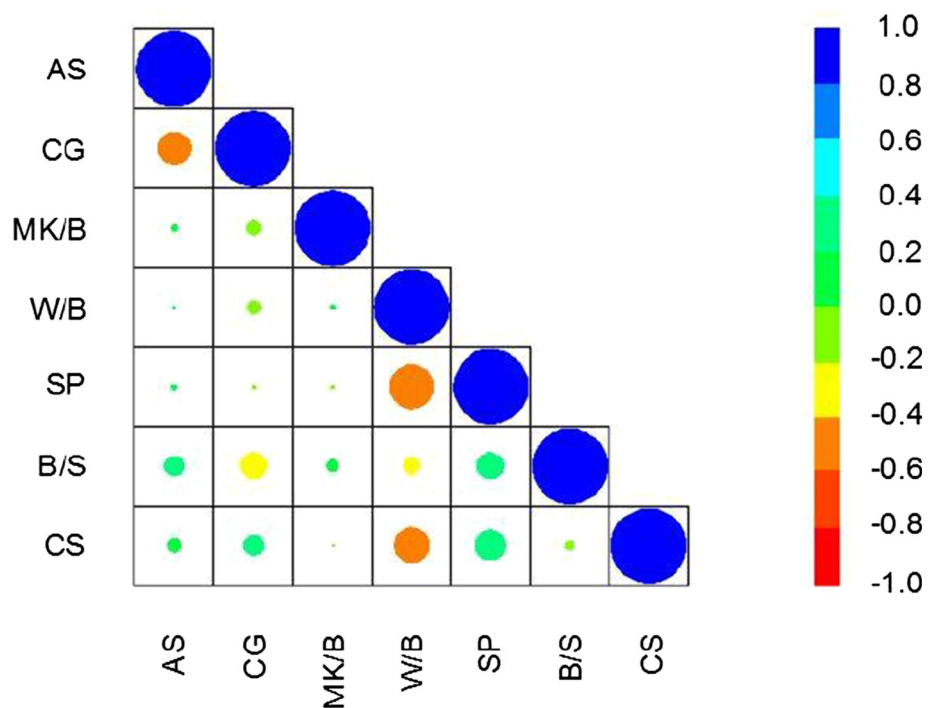
In addition, Figs. 5, 6 and 7 demonstrate the frequency histograms of the parameters used for the modeling of the cement-based mortar's CS. These figures are extremely useful, as they determine the range of parameter values, where data are insufficient. When on the one hand

Table 2 Input and output parameters used in the development of AI and SC techniques (all datasets)

No	Variable	Symbol	Units	Category	Statistics			
					Min	Average	Max	STD
1	Age of Specimen	AS	Days	Input	1	34	90	30.71
2	Cement Grade	CG	MPa	Input	32.00	39.98	53.00	9.65
3	Metakaolin percentage in relation to total binder	MK/B	(%w/w)	Input	0.00	14.39	30.00	10.15
4	Water-to-binder ratio	W/B	w/w	Input	0.36	0.47	0.60	0.06
5	Superplasticizer	SP	(%w/w)	Input	0.00	0.26	2.35	0.55
6	Binder-to-sand ratio	B/S	w/w	Input	0.33	0.45	0.50	0.07
7	Compressive strength	CS	MPa	Output	6.06	42.81	99.17	16.67

Table 3 Correlation matrix of the variables

Variables		Input						Output
		AS	CG	MK/B	W/B	SP	B/S	CS
Input	AS	1.00						
	CG	− 0.43	1.00					
	MK/B	0.07	− 0.18	1.00				
	W/B	0.02	− 0.16	0.05	1.00			
	SP	0.06	− 0.04	− 0.04	− 0.60	1.00		
	B/S	0.26	− 0.35	0.16	− 0.22	0.35	1.00	
Output	CS	0.17	0.26	− 0.03	− 0.47	0.39	− 0.11	1.00

Fig. 4 Correlogram of the variables (input and output parameters)

sufficient data exist, covering a wide range of parameters, then the model leads to a reliable forecast. On the other hand, if a certain range of an input parameter is insufficient

or even lacking, a less reliable forecast usually occurs, because the model does not have enough data to be trained. Furthermore, these figures highlight areas where

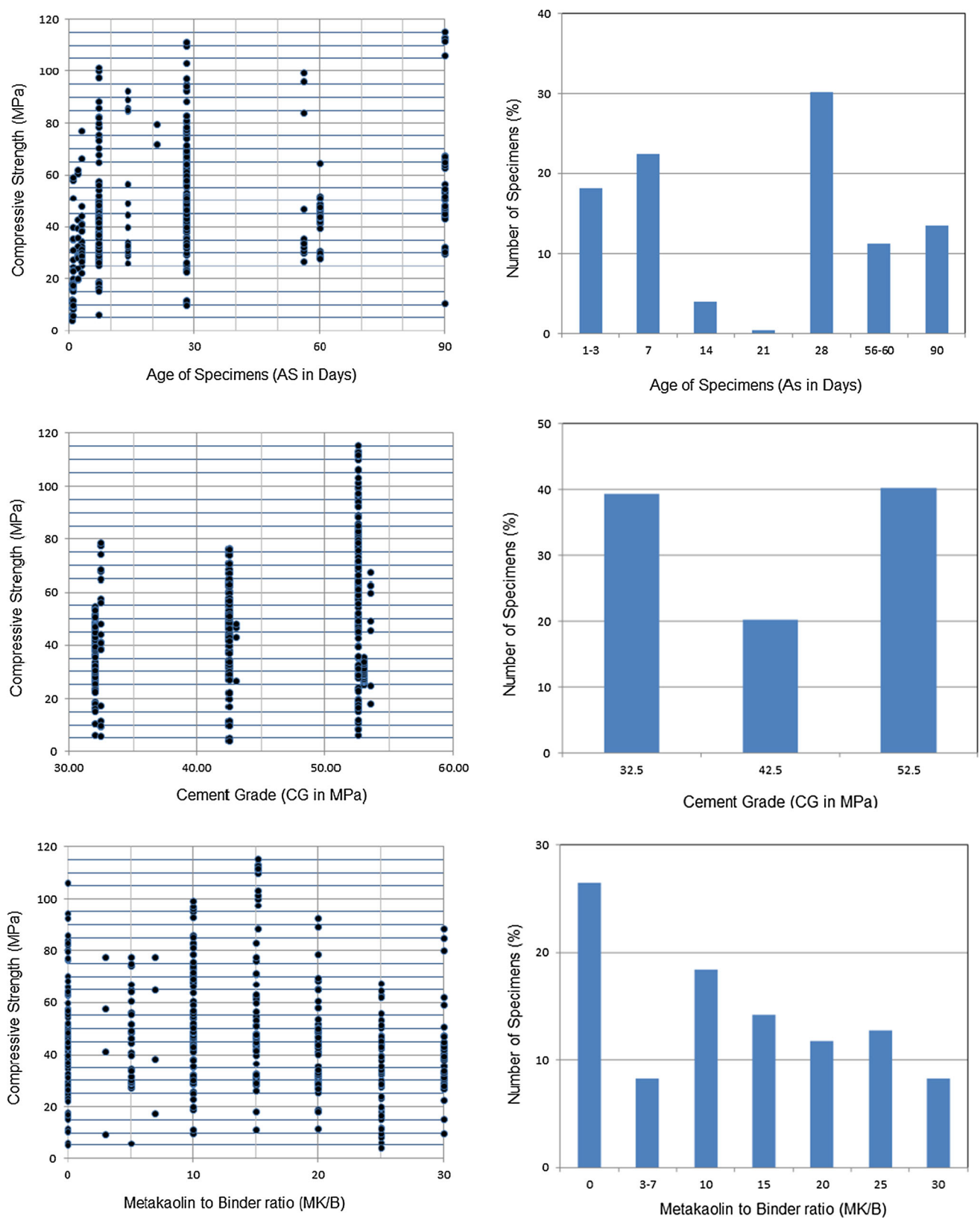


Fig. 5 Histograms of the parameters: age of specimen (AS), the cement grade (CG), the metakaolin percentage in relation to total binder (MK/B

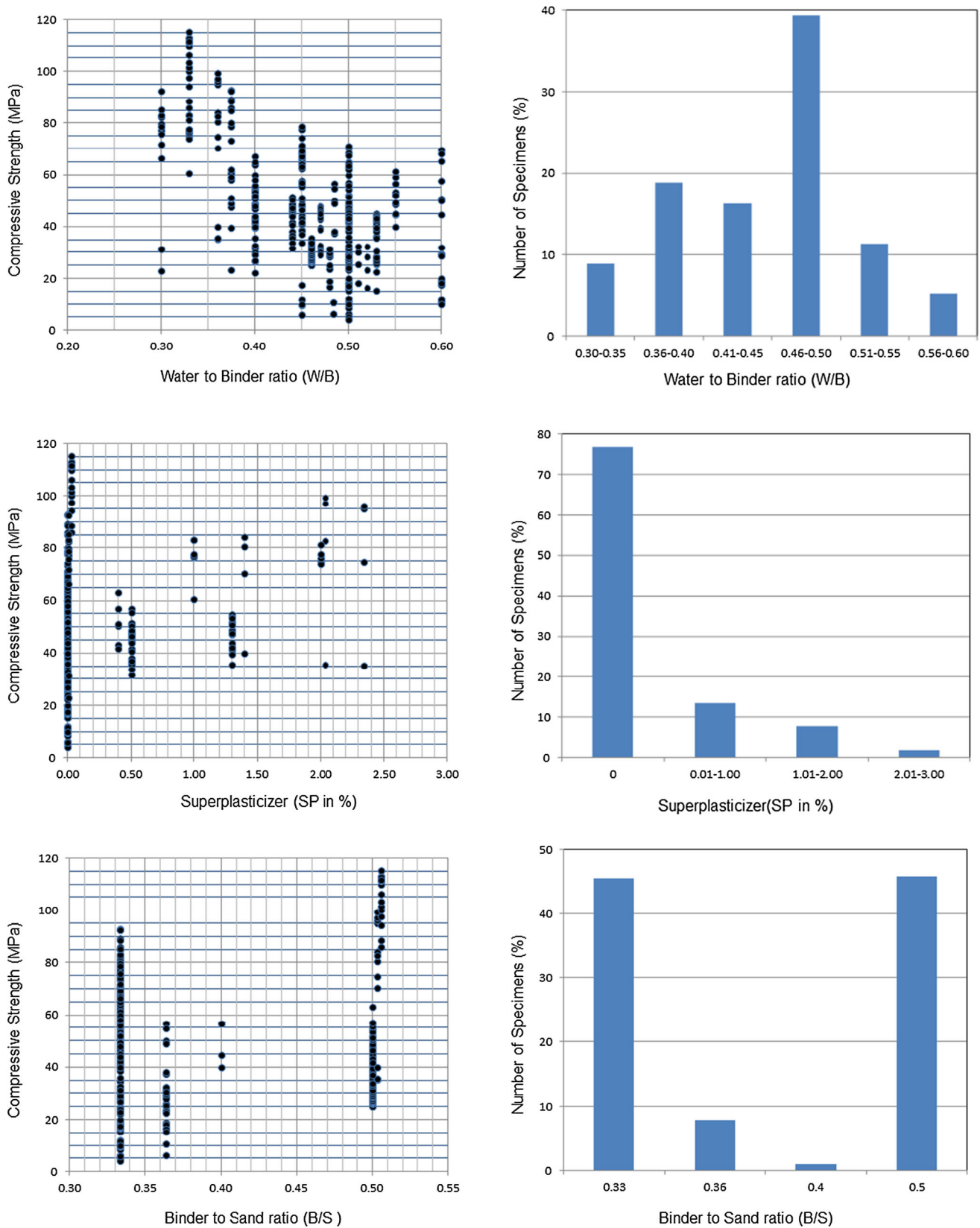


Fig. 6 Histograms of the parameters: Water-to-binder ratio (W/B), the superplasticizer (SP), which is the percentage of the addition of superplasticizer in relation to the total binder (%w/w), and the Binder-to-Sand ratio (B/S)

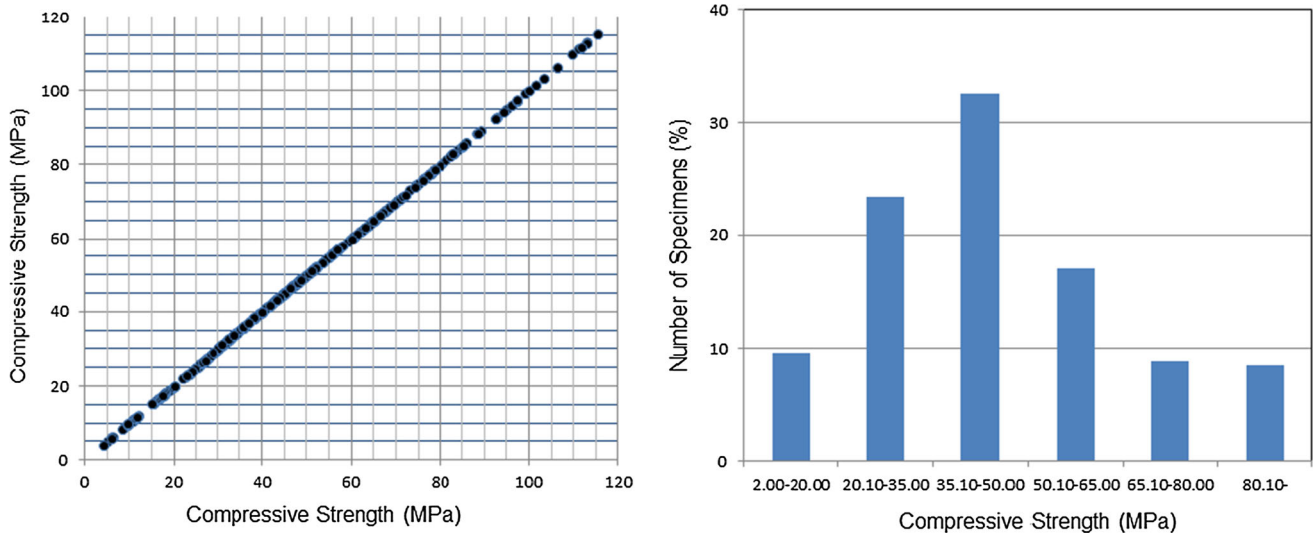


Fig. 7 Histograms of the parameters: compressive strength (CS),

experimental data are lacking and guide mortar mix designers to investigate the aforementioned synthesis, therefore, not only leading to new experimental results, but also enhancing the database itself.

3.3 Short review on the behavior of cement-based mortars

Herein many different works have been reviewed, each one of them containing a particular approach regarding the CS of specimens. There are similarities among those published works but there are also differences that provide an interesting approach on the matter that is discussed. Most researchers wanted to find solutions regarding the very heavy energy-consuming concrete and mortar production by reducing the cement's presence in the admixture without, at the same time, affecting excessively the mechanical properties, with the CS being this paper's focus. Some researchers like Vu et al. 2001 [1], wanted to significantly alter the mixtures by min-maxing each and every part of the mixture such as the water-to-binder ratio, the metakaolin percentage, or need for using a superplasticizer.

A very interesting experiment is given in Potgieter-Vermaak and Potgieter [84] where the optimum temperature required for the production of the most effective metakaolin is determined. Now, as it regards metakaolin the most prominent candidate for cement replacement according to all researchers, the conclusion was that roughly 20% is the most adequate substitution to the cement in order to enhance the specimens' CS. All researchers determined that over 30% of metakaolin in the binder has negative effects on the specimens' CS and should be avoided.

There are a number of researchers, like Ashok et al. [90] and Cyr et al. [88], who explore the usage of even more radical by-product materials as substitutes with promising results, such as nanosilica and nanoclay metakaolin or municipal solid waste incineration fly ash and sewage sludge ash. These should be further investigated in the future. There are also researchers who explored the possibility of using metakaolin in harsh environments, like Khater [92], Lee et al. [95], with results that might not be promising for usage in the production process as of now, but they have some satisfactory results to consider in different future experimental campaigns. Saidat et al. [89] gave another interesting dimension in the endeavor for metakaolin incorporation, by exploring the usage of different activators for metakaolin in order to enhance its properties.

3.4 Sensitivity analysis of the parameters affecting the CS of mortars based on experimental database

In general, sensitivity analysis (SA) of a numerical model is a technique used to determine whether the output of the model is affected by changes in the input parameters. This provides feedback as to which input parameters are the most significant, and thus, by removing the insignificant ones, the input space will be reduced and subsequently the complexity of the model as well as the time required for its training will be also reduced. In order to identify the effects of model inputs on the outputs, the SA can be conducted on the database. Sometimes, the results of SA helps researchers/designers to remove one or more input parameters from the database to obtain better analyses with a higher level of performance prediction. To perform the

SA, the cosine amplitude method (CAM) which has been used by many researchers [96–98] was selected and implemented. In CAM, data pairs are used to construct a data array, X , as follows:

$$X = \{x_1, x_2, x_3, \dots, x_i, \dots, x_n\} \quad (4)$$

Variable x_i in array, X , is a length vector of m as:

$$x_i = \{x_{i1}, x_{i2}, x_{i3}, \dots, x_{im}\} \quad (5)$$

The relationship between R_{ij} (strength of the relation) and datasets of X_i and X_j is presented by the following equation:

$$R_{ij} = \frac{\sum_{k=1}^m x_{ik}x_{jk}}{\sqrt{\sum_{k=1}^m x_{ik}^2 \sum_{k=1}^m x_{jk}^2}} \quad (6)$$

The R_{ij} values between the CS and the input parameters are shown in Fig. 8. This analysis revealed that the CG has the greatest influence on the CS values, with a R_{ij} value of 0.9107, followed by the B/S (0.8901), W/B (0.8580), AS (0.7173), MK/B (0.6803) and the parameter with the lowest influence on the CS seems to be the SP (0.4539).

4 Methodology and results

4.1 Performance indices

Three different statistical parameters were employed to evaluate the performance of the derived model as well as the available in the literature formulae, including the root-mean-square error (RMSE), the mean absolute percentage error (MAPE), and the coefficient of determination (R^2). Lower RMSE and MAPE values represent more accurate prediction results (a null value indicates a perfect fit), while

higher R^2 values represent a better fit between the analytical and predicted values (a unit value indicates a perfect fit). The aforementioned statistical parameters have been calculated by the following expressions [99]:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - y_i)^2} \quad (7)$$

$$\text{MAPE} = \frac{1}{n} \sum_{i=1}^n \left| \frac{x_i - y_i}{x_i} \right| \quad (8)$$

$$R^2 = 1 - \left(\frac{\sum_{i=1}^n (x_i - y_i)^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right) \quad (9)$$

where n denotes the total number of datasets, x_i and y_i represent the predicted and target values, respectively.

It should be highlighted that, among the statistical indices available, the majority of researchers use the R^2 , in order to evaluate the effectiveness of the developed computation model. According to the Cauchy–Schwarz inequality [100], the coefficient R has a value between $+1$ and -1 . The further away R is from zero, the stronger the linear relationship is between the two variables. The sign of R corresponds to the direction of the relationship. Often, the Pearson correlation coefficient is used to calculate the slope of the linear relationship between two variables, in accordance with the following linear expression:

$$X = aY + c \quad (10)$$

When two forecasting models present different R values, as well as different slope values, comparison between the models is impossible. Even more so, when evaluating neural networks developed through different architectures. Therefore, the following new engineering index, the a20-

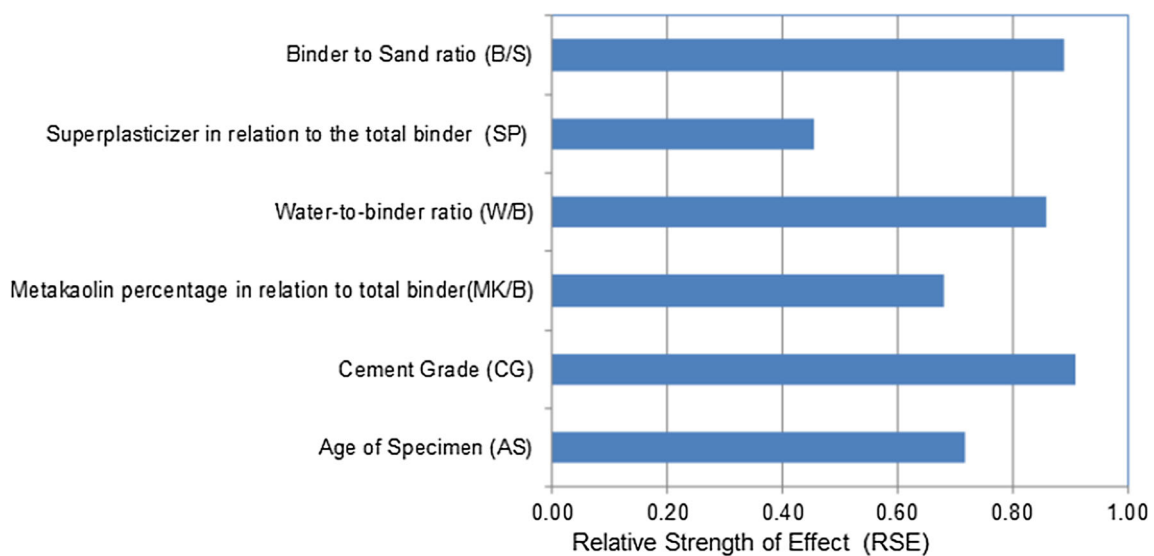


Fig. 8 Sensitivity analysis of cement–metakaolin mortars CS based on experimental database

index, has been recently proposed [12, 14, 18, 101–108] for the reliability assessment of the developed SC techniques:

$$a20 - \text{index} = \frac{m20}{M} \quad (11)$$

where M is the number of dataset sample and $m20$ is the number of samples with value of ratio experimental value/predicted value between 0.80 and 1.20. Note that for a perfect predictive model, the values of $a20$ -index values are expected to be the unit value. The proposed $a20$ -index has the advantage of a physical engineering meaning, as it declares the amount of the samples that satisfy the predicted values with a deviation of $\pm 20\%$ compared to experimental values.

4.2 Methodology adopted

: The aim of this paper is to use new techniques to solve complex and nonlinear problems. Since previous statistical methods are less flexible and have limitations, AI-based computing techniques have been considered. In this research, as stated, various methods were used, namely KNN, SVM, RF, DT and AdaBoost models. Since each AI method is based on a specific structure and parameters, it can offer different functions to solve problems. Therefore, various structures were developed in this study to obtain predictive models for cement-based mortars CS. To increase the performance of the models, their structures are analyzed and their parameters are identified. To compare the performance of the models, four types of statistical indices were used, namely MAPE, RMSE, R^2 and $a20$ -index. These indices can assess the quality of the models with good accuracy and make the comparison between them possible.

The structural design of each model has characteristics evaluated next. The data used in SC and AI models are firstly analyzed, and then the data that are not strongly related to other data are discarded. This process of model training and data correction continues, by trial and error, so that the best quality of each model can be obtained to predict cement-based mortars' CS. If the required criteria are confirmed, the test data are given to the models to analyze their performance against the new conditions.

4.3 Results and discussion

As stated, five AI models were developed to predict cement-based mortars' CS. According to the data described above, 424 datasets were collected and used for simulation. Eighty percent of the data were used to design the models, and 20% of the data were used to evaluate the performance of the models. The data were randomly divided and parametric analysis was performed to obtain the best model,

presented in Table 4, which gathers the optimal features of each designed model.

Given that five models of AI were used for prediction, statistical indices were used to evaluate their performance. Table 5 presents the results of the models developed for cement-based mortars' CS. As it can be seen, the performance of each model reflects its capabilities. In general, the error of models designed based on the tree base model (RF and AdaBoost models) is lower than other models. The R^2 of RF and AdaBoost models are 0.9768 and 0.9965 for the training data, respectively. Subsequent evaluation with test data shows that the best performance was, again, provided by these models with values of R^2 of RF and AdaBoost models equal to 0.9439 and 0.9473, respectively. This suggests that the simulation was performed with acceptable accuracy and has good flexibility for new data. Finally, a comparison between the models shows that the $a20$ -index reports the best results for the AdaBoost model. Values of 0.985 and 0.905 for the AdaBoost model are obtained for the training and testing data, respectively. The new $a20$ -index criterion, among other performance indices, can have a better distribution of the results near to $Y = X$ line. It can be used to prevent computational errors for cement-based mortars' CS evaluation. Finally, the results of the five AI models are used to predict cement-based mortars' CS for training and testing data in Figs. 9, 10, 11, 12, 13, 14, 15, 16, 17 and 18.

Considering the importance of predicting cement-based mortars' CS, the SA was performed based on the two better models. This analysis reports the effect of changes in input parameters on the output parameter, based on the RF and AdaBoost models. Figure 19 and Fig. 20 show the results of this SA for RF and AdaBoost models, respectively. As it can be seen, each parameter has changed from 0 to 100%. By changing each parameter, the other parameters are kept constant so that the effects can be correctly seen. The results show that the changes in the two parameters B/S and W/B indicate the largest effect after roughly 40% and 30% of the changes, respectively. In addition, we can mention the trend of changes in these two models, which almost explain the close trends that confirm the accuracy of the models. By examining the initial parameters in laboratory samples and in these two developed models, it can be seen that the two parameters of B/S and W/B have the most effects on the CS values and in fact, confirm that the models are properly developed.

5 Conclusions

This research presents a novel and alternative method for the evaluation of the compressive strength (CS) of mortar specimens using artificial intelligence (AI) models. For this

Table 4 Description of developed models in predicting cement-based mortars' CS values

Models	Parameters	Optimum parameters
KNN	Number of neighbors	3
	Function type	Minkowski
SVM	SVM type	v-SVM
	Kernel	Sigmoid
Random forest	Number of trees	10
	Split subset limit	> 5
Decision tree	Min. number of instances in leaves	2
	Split subset limit	> 6
	Max. tree depth	5
AdaBoost	Base estimator	Tree
	Number of estimators	30

Table 5 Obtained results of developed models

Section	Model	MAPE	RMSE	R^2	a20 – index
Training	KNN	0.169	7.527	0.8807	0.859
	SVM	0.458	17.736	0.3786	0.368
	RF	0.074	3.359	0.9768	0.950
	DT	0.218	9.256	0.8186	0.794
	AdaBoost	0.017	1.305	0.9965	0.985
Testing	KNN	0.143	7.789	0.8740	0.845
	SVM	0.434	17.667	0.4203	0.393
	RF	0.109	5.240	0.9439	0.893
	DT	0.189	8.439	0.8526	0.774
	AdaBoost	0.097	5.291	0.9473	0.905

purpose, 424 datasets were from the literature for the estimation of the cement-based mortars' CS. The reported database contains the following six input parameters for the prediction of the CS, i.e., age of specimens, cement grade, metakaolin percentage in relation to total binder, the water-to-binder ratio, the superplasticizer percentage in relation to the total binder, and the binder-to-sand ratio. After a preliminary investigation of the available AI and soft computing (SC) techniques, SVM, RF, DT, KNN and AdaBoost were used to estimate the compressive strength

of cement-based mortars. Each of these techniques has its own effective parameters and their optimal values needs to be determined. The results show that the highest prediction accuracy was obtained from the AdaBoost (Adaptive Boosting) and RF (Random Forest) models. The models were ranked using four performance indices, i.e., R^2 , RMSE, MAPE and a20-index and comparing them for each predictive model. The performance indexes' results based on R^2 were obtained as (0.9965 and 0.9768) and (0.9473 and 0.9439) for training and testing phases of the AdaBoost and RF models, respectively. Finally, a sensitivity analysis of these models showed very good confidence in the prediction accuracy of the compressive strength of cement-based mortars.

The modeling process and results of the present study can be used by researchers, designers and engineers interested in predicting the strength of cement-based mortars and in developing new eco-efficient compositions for a wide application in civil engineering. One of the limitations of the developed models in this study is to obtain the optimal structure using different analyses. Therefore, it is suggested to optimize the coefficients/weights of the developed models using some metaheuristic optimization algorithms. In addition, a bigger database of cement-based materials' mechanical characteristics can be provided in order to develop another new SC technique.

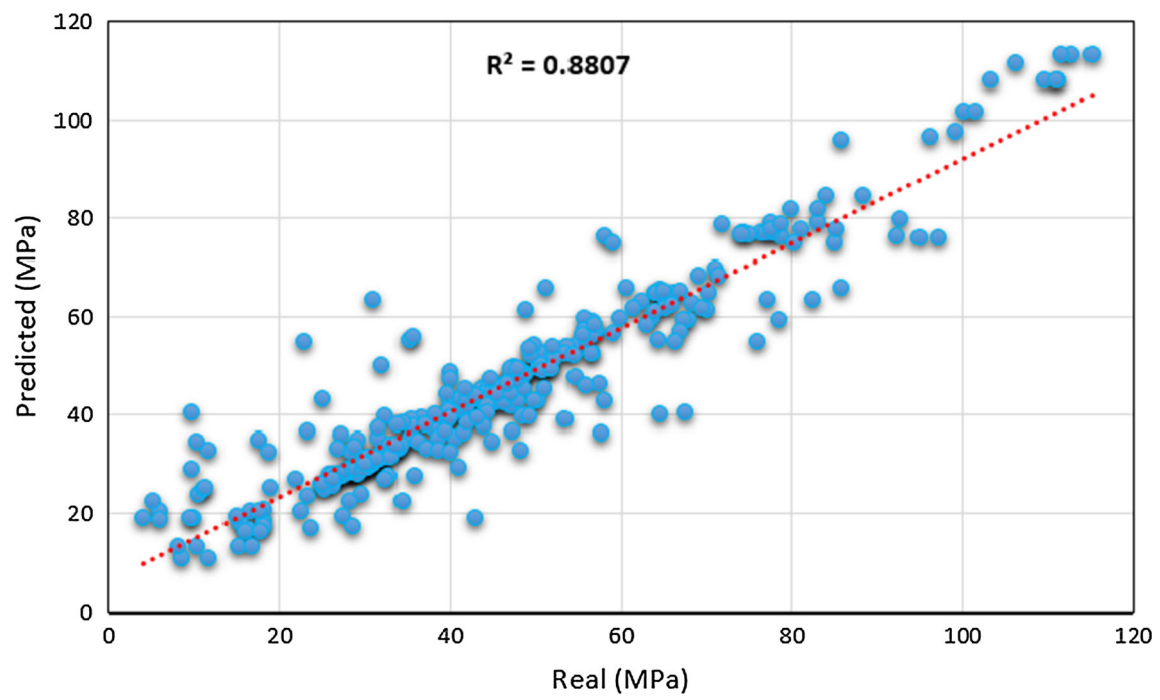


Fig. 9 Result of training data for KNN model

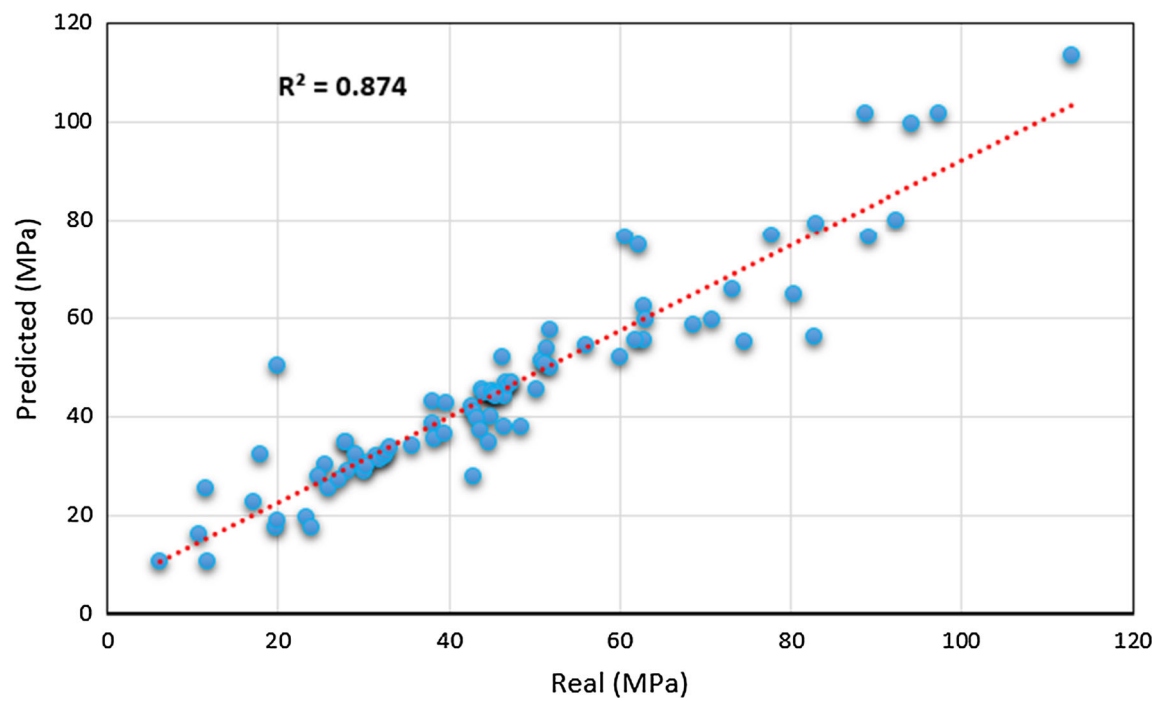


Fig. 10 Result of testing data for KNN model

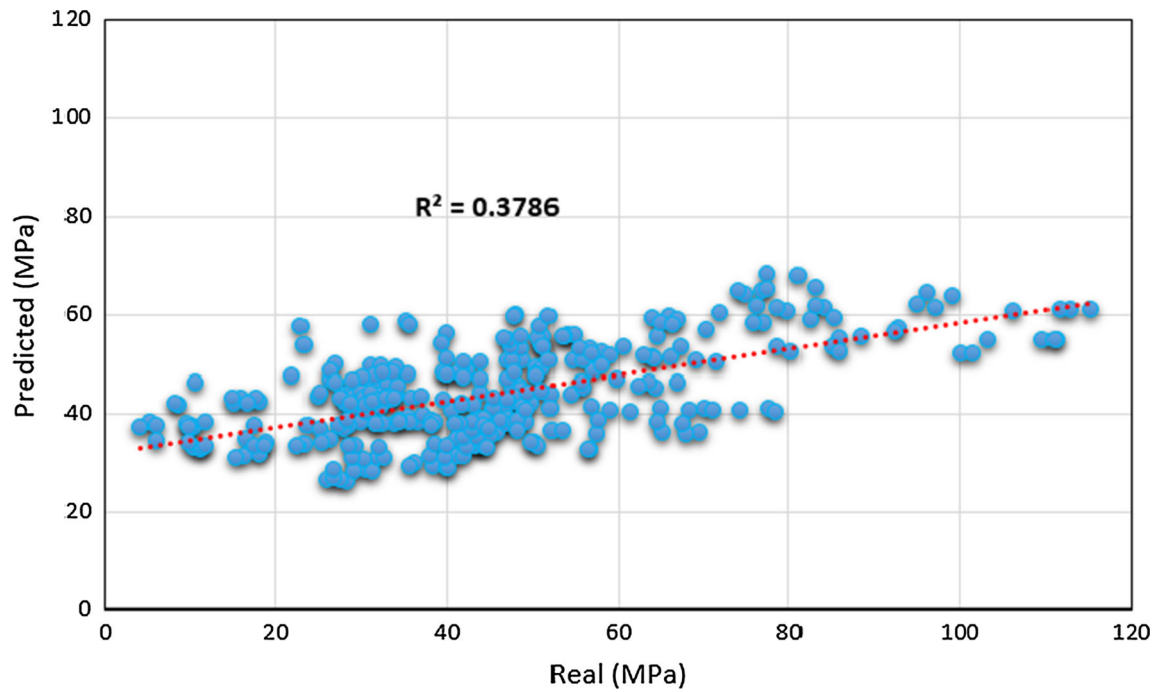


Fig. 11 Result of training data for SVM model

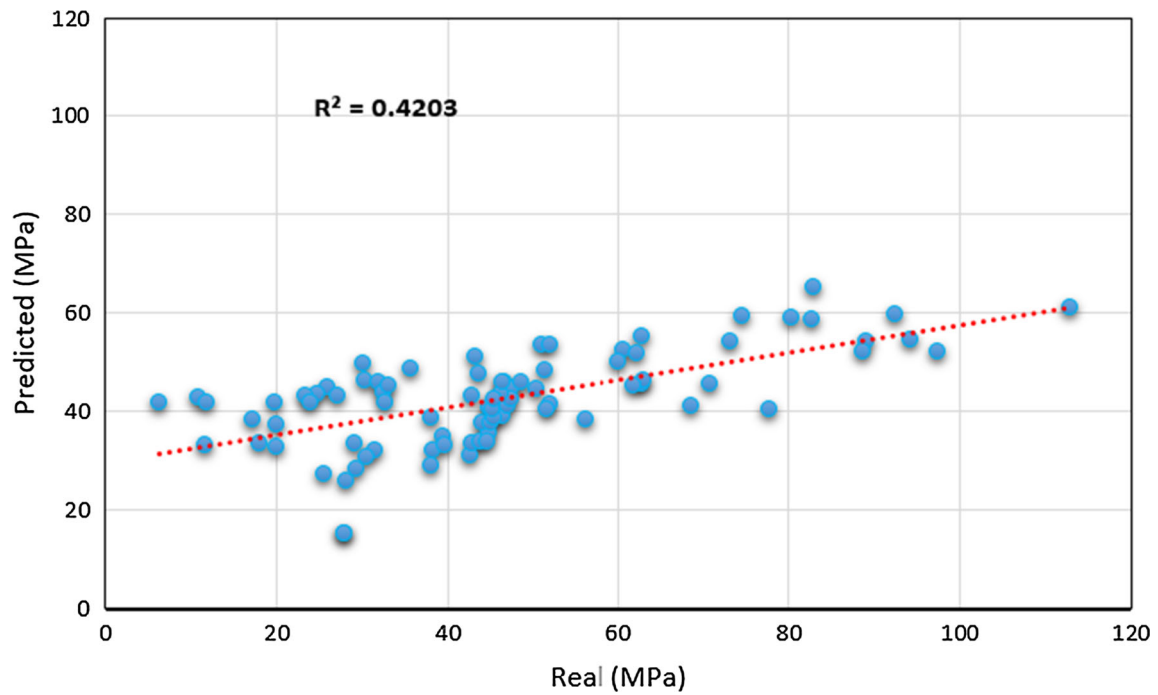


Fig. 12 Result of testing data for SVM model

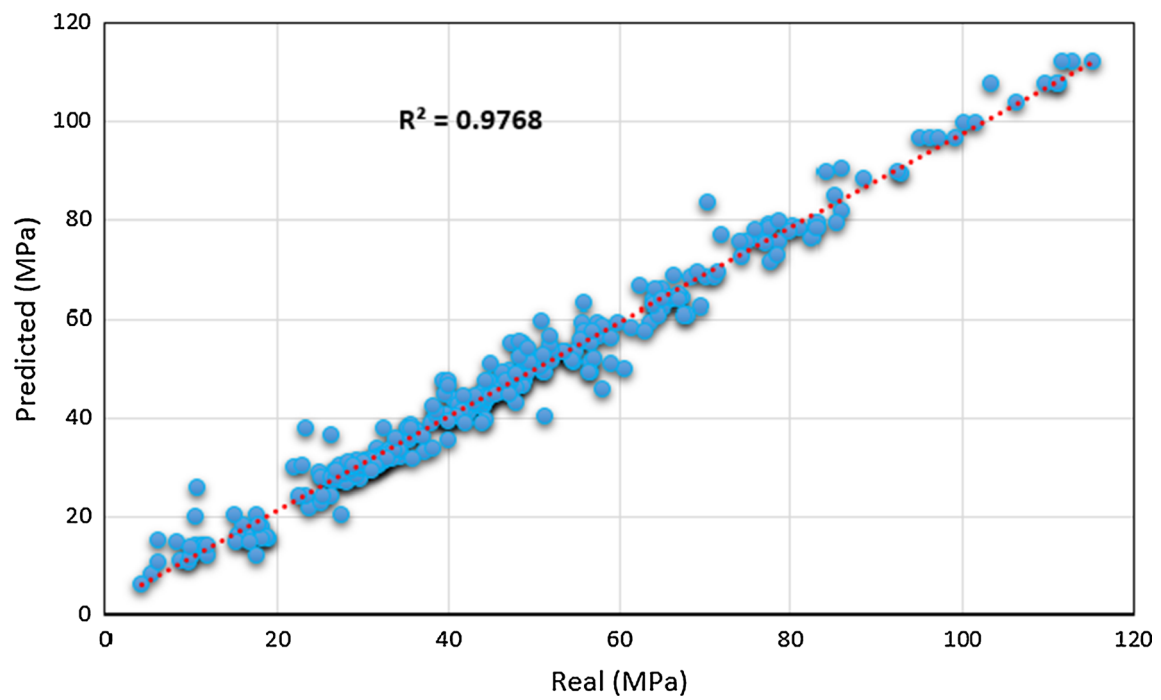


Fig. 13 Result of training data for RF model

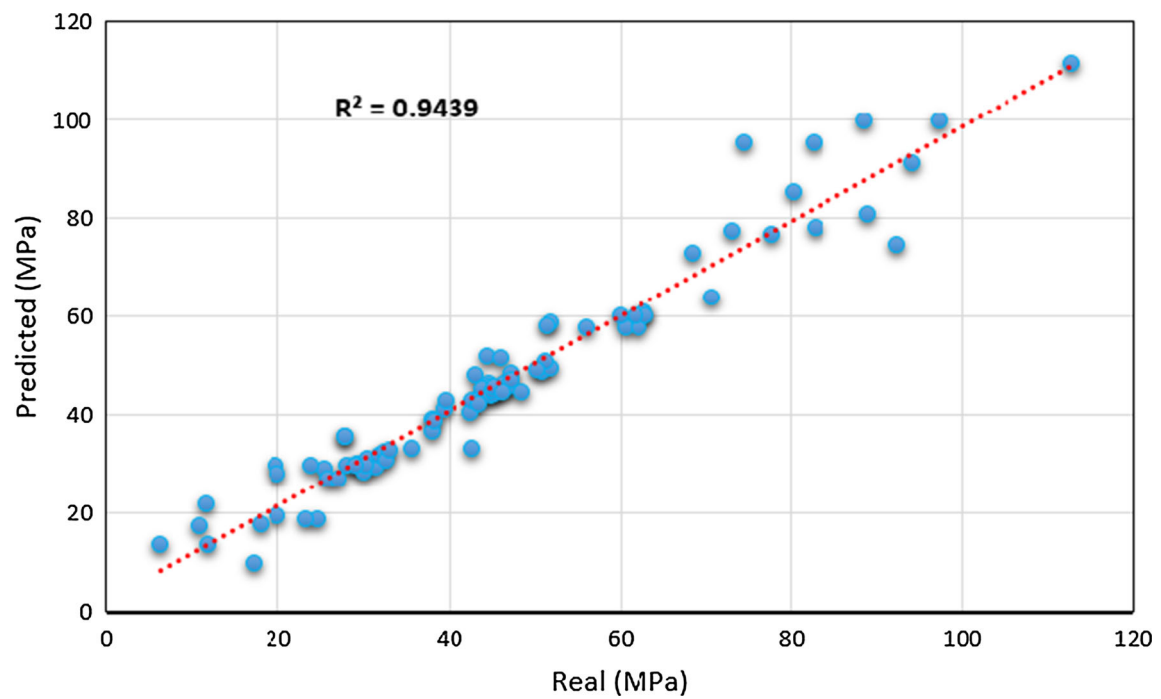


Fig. 14 Result of testing data for RF model

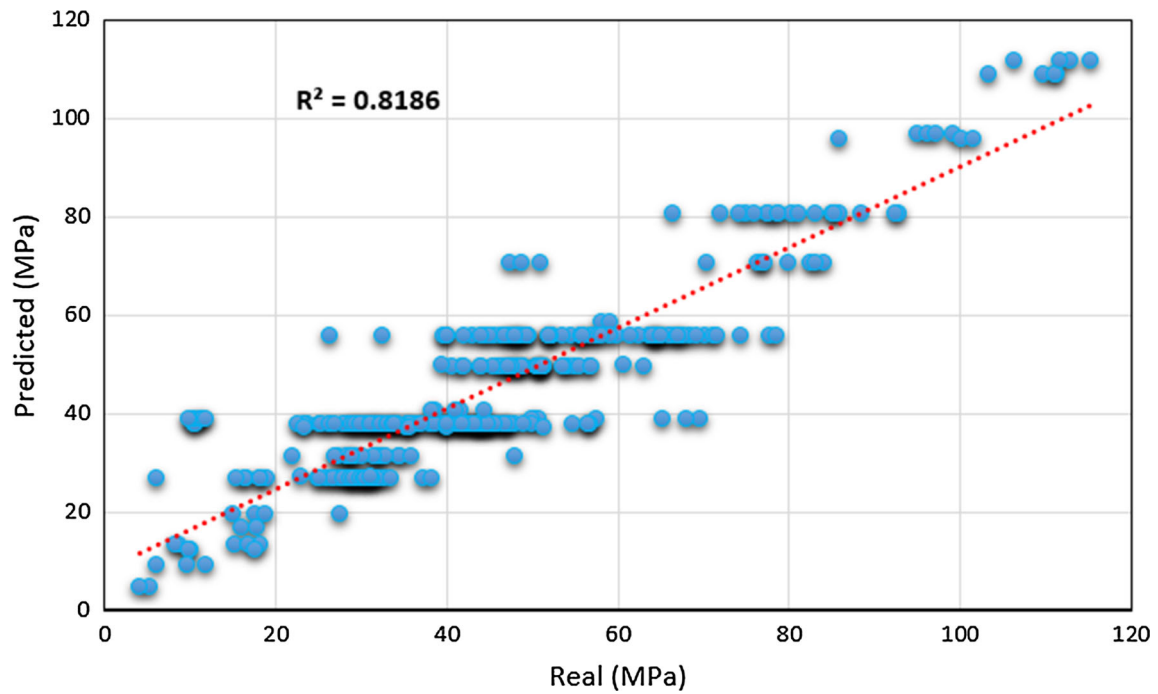


Fig. 15 Result of training data for DT model

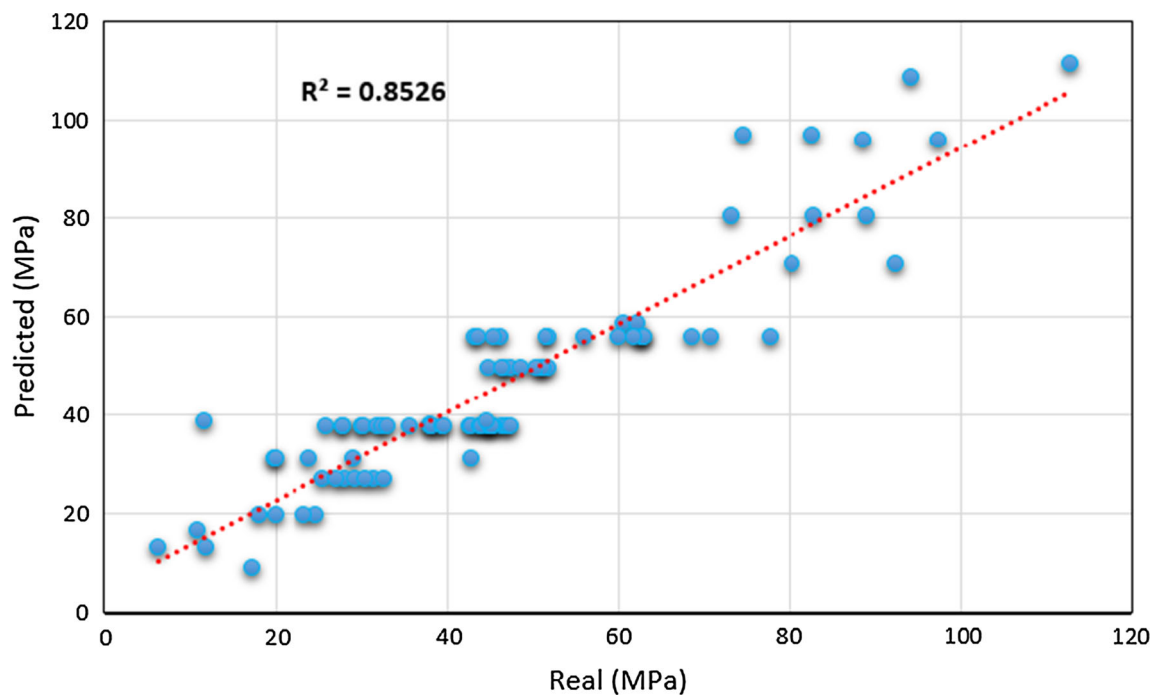


Fig. 16 Result of testing data for DT model

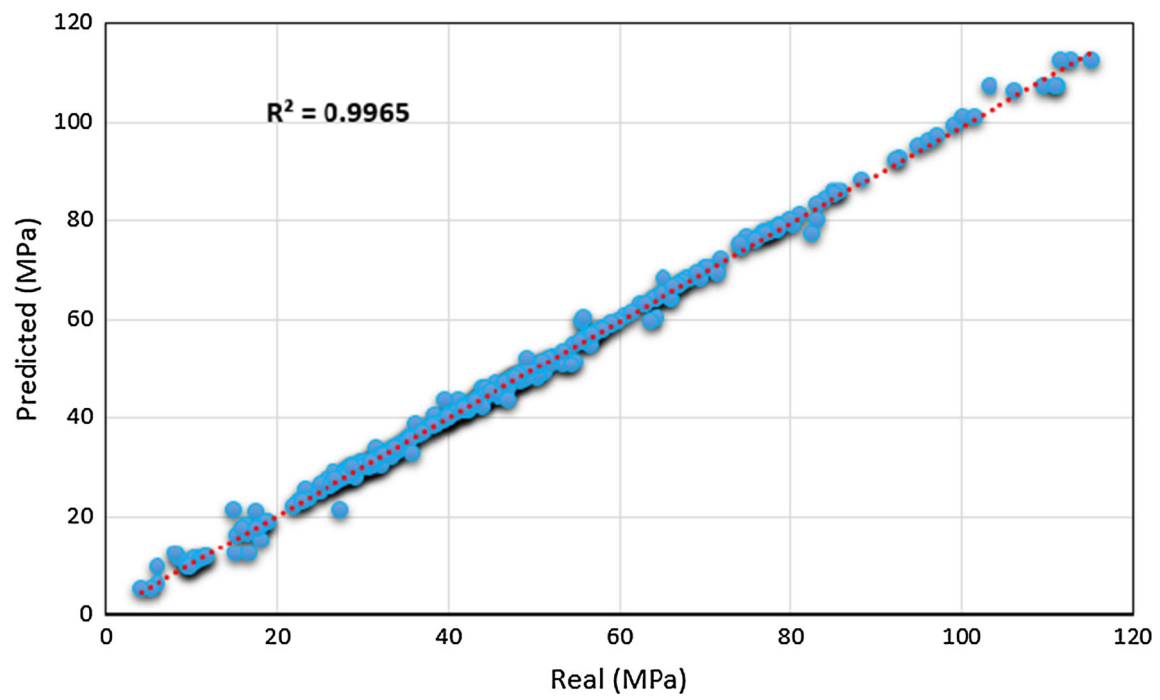


Fig. 17 Result of training data for AdaBoost model

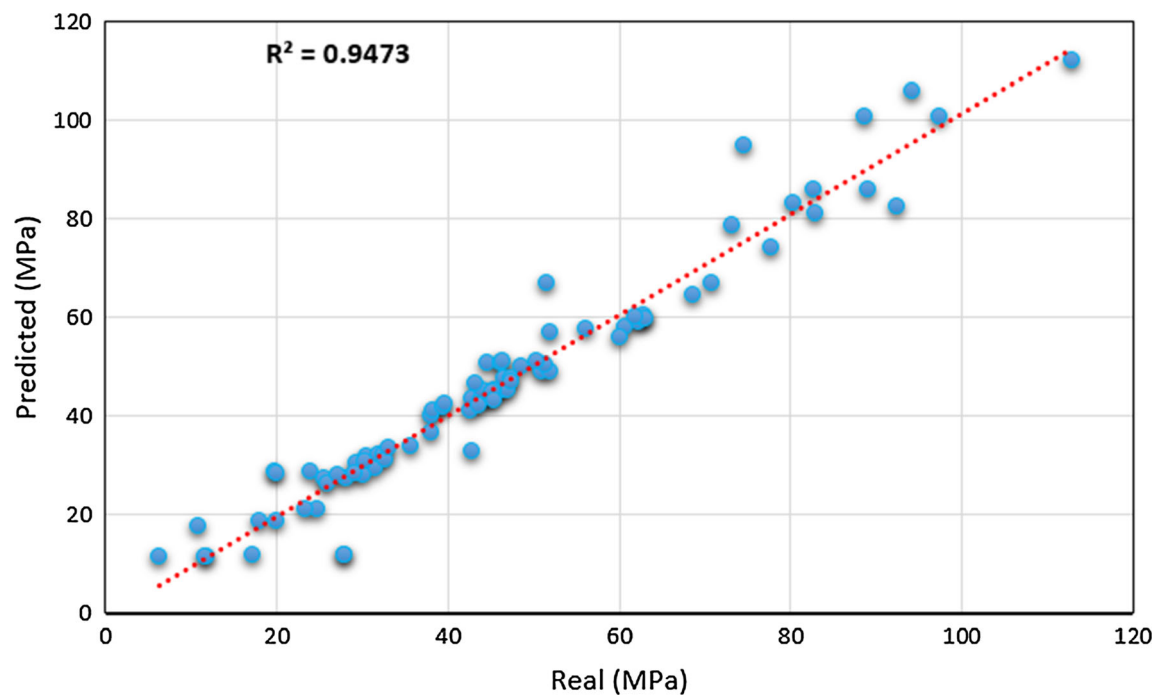


Fig. 18 Result of testing data for AdaBoost model

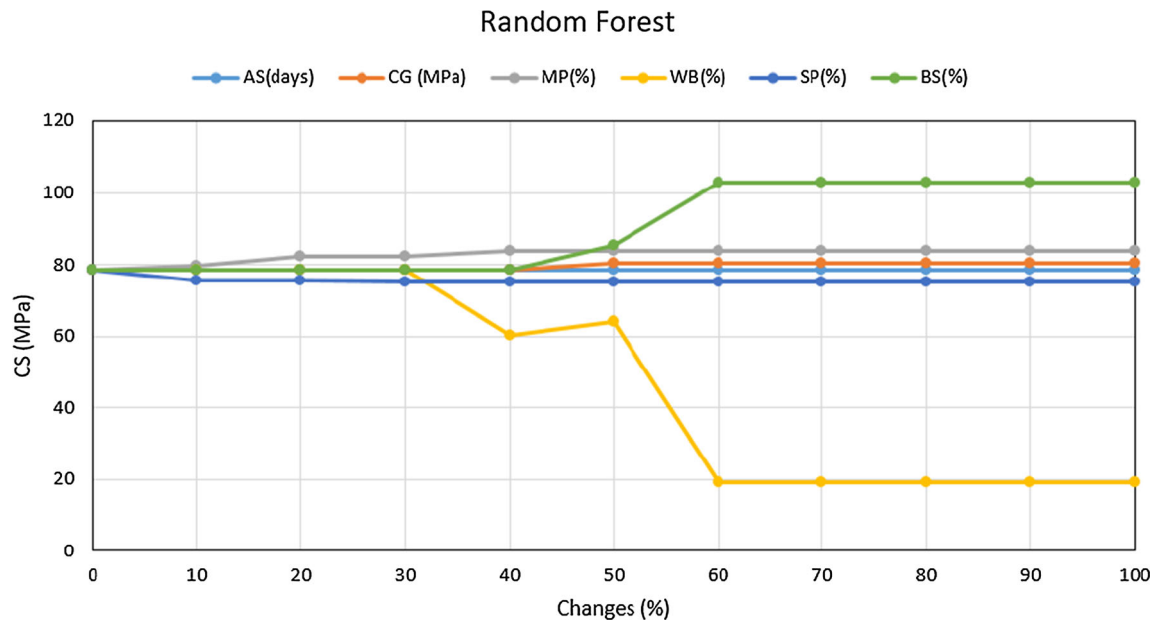


Fig. 19 Changes of parameter on RF model

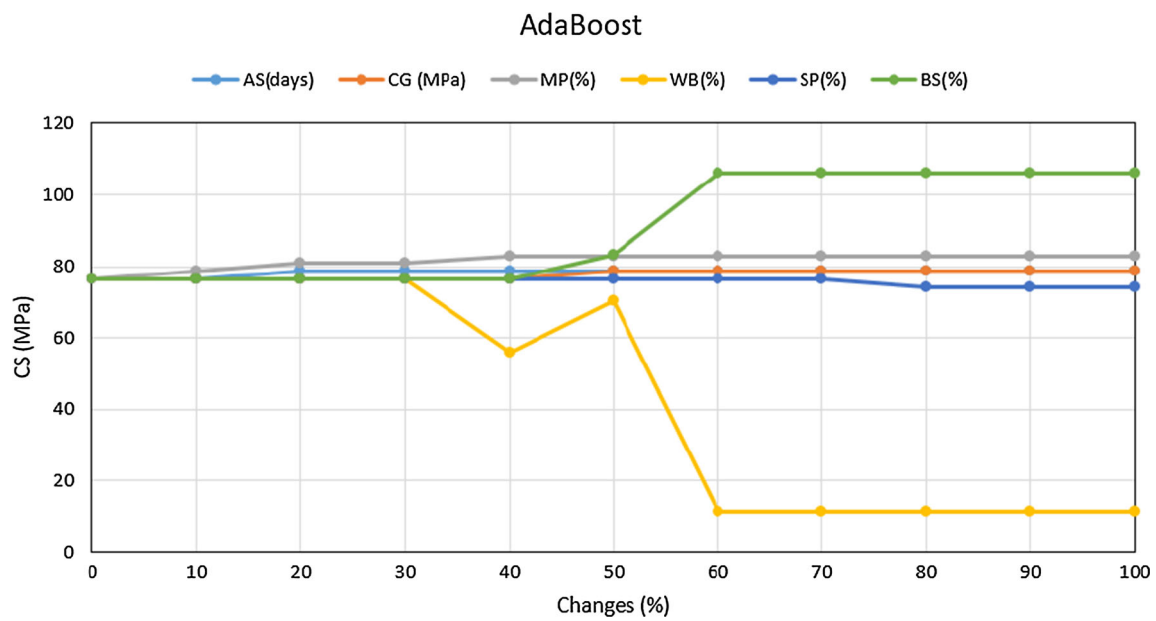


Fig. 20 Changes of parameter on AdaBoost model

Appendix

See Table 6.

Table 6 Cement-based mortars experimental database

No		Sample Code	Input Parameters							Output Parameter	References
Per database	Per work		AS	CG	MK/B	W/B	SP	B/S	CS		
-	-	-	(days)	MPa	(%w/w)	(w/w)	(%w/w)	(w/w)	MPa	-	-
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	
1	1	M1	7	32.00	0.00	0.48	0.00	0.36	16.56	Vu et al. 2001 [1]	
2	2	M2	7	32.00	10.00	0.48	0.00	0.36	18.99		
3	3	M3	7	32.00	15.00	0.50	0.00	0.36	18.17		
4	4	M4	7	32.00	20.00	0.51	0.00	0.36	18.13		
5	5	M5	7	32.00	25.00	0.52	0.00	0.36	16.30		
6	6	M6	7	32.00	30.00	0.53	0.00	0.36	15.45		
7	7	M7	7	32.00	0.00	0.53	0.00	0.50	25.50		
8	8	M8	7	32.00	10.00	0.53	0.00	0.50	25.90		
9	9	M9	7	32.00	15.00	0.53	0.00	0.50	28.40		
10	10	M10	7	32.00	20.00	0.53	0.00	0.50	28.20		
11	11	M11	7	32.00	25.00	0.53	0.00	0.50	27.50		
12	12	M12	7	32.00	30.00	0.53	0.00	0.50	26.80		
13	13	M13	7	32.00	0.00	0.50	0.00	0.50	29.30		
14	14	M14	7	32.00	10.00	0.50	0.00	0.50	30.40		
15	15	M15	7	32.00	15.00	0.50	0.00	0.50	31.30		
16	16	M16	7	32.00	20.00	0.50	0.00	0.50	29.40		
17	17	M17	7	32.00	25.00	0.50	0.00	0.50	29.10		
18	18	M18	7	32.00	30.00	0.50	0.00	0.50	26.80		
19	19	M19	7	32.00	0.00	0.47	0.00	0.50	31.50		
20	20	M20	7	32.00	10.00	0.47	0.00	0.50	31.90		
21	21	M21	7	32.00	15.00	0.47	0.00	0.50	32.60		
22	22	M22	7	32.00	20.00	0.47	0.00	0.50	30.40		
23	23	M23	7	32.00	25.00	0.47	0.00	0.50	30.20		
24	24	M24	7	32.00	30.00	0.47	0.00	0.50	29.00		
25	25	M25	7	32.00	0.00	0.44	0.50	0.50	35.40		
26	26	M26	7	32.00	10.00	0.44	0.50	0.50	38.00		
27	27	M27	7	32.00	15.00	0.44	0.50	0.50	36.60		
28	28	M28	7	32.00	20.00	0.44	0.50	0.50	35.70		
29	29	M29	7	32.00	25.00	0.44	0.50	0.50	33.80		
30	30	M30	7	32.00	30.00	0.44	0.50	0.50	31.70		
31	31	M31	7	32.00	0.00	0.40	1.30	0.50	40.00		
32	32	M32	7	32.00	10.00	0.40	1.30	0.50	42.40		
33	33	M33	7	32.00	15.00	0.40	1.30	0.50	41.90		
34	34	M34	7	32.00	20.00	0.40	1.30	0.50	41.40		
35	35	M35	7	32.00	25.00	0.40	1.30	0.50	39.60		
36	36	M36	7	32.00	30.00	0.40	1.30	0.50	35.50		
37	37	M1	28	32.00	0.00	0.48	0.00	0.36	23.68		
38	38	M2	28	32.00	10.00	0.48	0.00	0.36	25.19		
39	39	M3	28	32.00	15.00	0.50	0.00	0.36	26.24		
40	40	M4	28	32.00	20.00	0.51	0.00	0.36	25.42		
41	41	M5	28	32.00	25.00	0.52	0.00	0.36	23.25		
42	42	M6	28	32.00	30.00	0.53	0.00	0.36	22.48		
43	43	M7	28	32.00	0.00	0.53	0.00	0.50	36.20		

Table 6 (continued)

No		Sample Code	Input Parameters						Output Parameter	References
Per database	Per work		AS	CG	MK/B	W/B	SP	B/S	CS	
-	-	-	(days)	MPa	(%w/w)	(w/w)	(%w/w)	(w/w)	MPa	-
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
44	44	M8	28	32.00	10.00	0.53	0.00	0.50	38.50	
45	45	M9	28	32.00	15.00	0.53	0.00	0.50	40.00	
46	46	M10	28	32.00	20.00	0.53	0.00	0.50	40.20	
47	47	M11	28	32.00	25.00	0.53	0.00	0.50	38.10	
48	48	M12	28	32.00	30.00	0.53	0.00	0.50	35.70	
49	49	M13	28	32.00	0.00	0.50	0.00	0.50	38.30	
50	50	M14	28	32.00	10.00	0.50	0.00	0.50	41.30	
51	51	M15	28	32.00	15.00	0.50	0.00	0.50	42.10	
52	52	M16	28	32.00	20.00	0.50	0.00	0.50	42.70	
53	53	M17	28	32.00	25.00	0.50	0.00	0.50	39.90	
54	54	M18	28	32.00	30.00	0.50	0.00	0.50	38.10	
55	55	M19	28	32.00	0.00	0.47	0.00	0.50	39.50	
56	56	M20	28	32.00	10.00	0.47	0.00	0.50	43.60	
57	57	M21	28	32.00	15.00	0.47	0.00	0.50	44.10	
58	58	M22	28	32.00	20.00	0.47	0.00	0.50	44.00	
59	59	M23	28	32.00	25.00	0.47	0.00	0.50	42.80	
60	60	M24	28	32.00	30.00	0.47	0.00	0.50	38.70	
61	61	M25	28	32.00	0.00	0.44	0.50	0.50	41.80	
62	62	M26	28	32.00	10.00	0.44	0.50	0.50	45.90	
63	63	M27	28	32.00	15.00	0.44	0.50	0.50	46.30	
64	64	M28	28	32.00	20.00	0.44	0.50	0.50	45.40	
65	65	M29	28	32.00	25.00	0.44	0.50	0.50	44.70	
66	66	M30	28	32.00	30.00	0.44	0.50	0.50	40.60	
67	67	M31	28	32.00	0.00	0.40	1.30	0.50	44.00	
68	68	M32	28	32.00	10.00	0.40	1.30	0.50	47.20	
69	69	M33	28	32.00	15.00	0.40	1.30	0.50	48.10	
70	70	M34	28	32.00	20.00	0.40	1.30	0.50	49.00	
71	71	M35	28	32.00	25.00	0.40	1.30	0.50	47.70	
72	72	M36	28	32.00	30.00	0.40	1.30	0.50	42.10	
73	73	M1	60	32.00	0.00	0.48	0.00	0.36	28.16	
74	74	M2	60	32.00	10.00	0.48	0.00	0.36	29.20	
75	75	M3	60	32.00	15.00	0.50	0.00	0.36	30.94	
76	76	M4	60	32.00	20.00	0.51	0.00	0.36	29.98	
77	77	M5	60	32.00	25.00	0.52	0.00	0.36	28.45	
78	78	M6	60	32.00	30.00	0.53	0.00	0.36	27.97	
79	79	M7	60	32.00	0.00	0.53	0.00	0.50	41.30	
80	80	M8	60	32.00	10.00	0.53	0.00	0.50	43.80	
81	81	M9	60	32.00	15.00	0.53	0.00	0.50	43.40	
82	82	M10	60	32.00	20.00	0.53	0.00	0.50	44.30	
83	83	M11	60	32.00	25.00	0.53	0.00	0.50	42.50	
84	84	M12	60	32.00	30.00	0.53	0.00	0.50	39.60	
85	85	M13	60	32.00	0.00	0.50	0.00	0.50	44.50	
86	86	M14	60	32.00	10.00	0.50	0.00	0.50	46.00	

Table 6 (continued)

No		Sample Code	Input Parameters						Output Parameter	References
Per database	Per work		AS (days)	CG MPa	MK/B (%w/ w)	W/B (w/ w)	SP (%w/ w)	B/S (w/ w)	CS MPa	
-	-	-	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
87	87	M15	60	32.00	15.00	0.50	0.00	0.50	44.70	
88	88	M16	60	32.00	20.00	0.50	0.00	0.50	45.10	
89	89	M17	60	32.00	25.00	0.50	0.00	0.50	43.60	
90	90	M18	60	32.00	30.00	0.50	0.00	0.50	42.10	
91	91	M19	60	32.00	0.00	0.47	0.00	0.50	46.30	
92	92	M20	60	32.00	10.00	0.47	0.00	0.50	46.90	
93	93	M21	60	32.00	15.00	0.47	0.00	0.50	45.20	
94	94	M22	60	32.00	20.00	0.47	0.00	0.50	47.00	
95	95	M23	60	32.00	25.00	0.47	0.00	0.50	43.90	
96	96	M24	60	32.00	30.00	0.47	0.00	0.50	43.60	
97	97	M25	60	32.00	0.00	0.44	0.50	0.50	46.60	
98	98	M26	60	32.00	10.00	0.44	0.50	0.50	48.50	
99	99	M27	60	32.00	15.00	0.44	0.50	0.50	47.80	
100	100	M28	60	32.00	20.00	0.44	0.50	0.50	48.70	
101	101	M29	60	32.00	25.00	0.44	0.50	0.50	47.30	
102	102	M30	60	32.00	30.00	0.44	0.50	0.50	43.90	
103	103	M31	60	32.00	0.00	0.40	1.30	0.50	48.70	
104	104	M32	60	32.00	10.00	0.40	1.30	0.50	50.90	
105	105	M33	60	32.00	15.00	0.40	1.30	0.50	51.10	
106	106	M34	60	32.00	20.00	0.40	1.30	0.50	52.00	
107	107	M35	60	32.00	25.00	0.40	1.30	0.50	51.30	
108	108	M36	60	32.00	30.00	0.40	1.30	0.50	47.50	
109	109	M1	90	32.00	0.00	0.48	0.00	0.36	29.74	
110	110	M2	90	32.00	10.00	0.48	0.00	0.36	31.23	
111	111	M3	90	32.00	15.00	0.50	0.00	0.36	32.10	
112	112	M4	90	32.00	20.00	0.51	0.00	0.36	32.16	
113	113	M5	90	32.00	25.00	0.52	0.00	0.36	32.28	
114	114	M6	90	32.00	30.00	0.53	0.00	0.36	30.40	
115	115	M7	90	32.00	0.00	0.53	0.00	0.50	43.70	
116	116	M8	90	32.00	10.00	0.53	0.00	0.50	44.50	
117	117	M9	90	32.00	15.00	0.53	0.00	0.50	45.00	
118	118	M10	90	32.00	20.00	0.53	0.00	0.50	44.20	
119	119	M11	90	32.00	25.00	0.53	0.00	0.50	43.40	
120	120	M12	90	32.00	30.00	0.53	0.00	0.50	43.00	
121	121	M13	90	32.00	0.00	0.50	0.00	0.50	47.00	
122	122	M14	90	32.00	10.00	0.50	0.00	0.50	46.70	
123	123	M15	90	32.00	15.00	0.50	0.00	0.50	46.90	
124	124	M16	90	32.00	20.00	0.50	0.00	0.50	45.30	
125	125	M17	90	32.00	25.00	0.50	0.00	0.50	45.30	
126	126	M18	90	32.00	30.00	0.50	0.00	0.50	44.90	
127	127	M19	90	32.00	0.00	0.47	0.00	0.50	48.00	
128	128	M20	90	32.00	10.00	0.47	0.00	0.50	47.40	
129	129	M21	90	32.00	15.00	0.47	0.00	0.50	47.40	

Table 6 (continued)

No		Sample Code	Input Parameters						Output Parameter CS	References
Per database	Per work		AS	CG	MK/B	W/B	SP	B/S		
-	-	-	(days)	MPa	(%w/w)	(w/w)	(%w/w)	(w/w)	MPa	-
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
130	130	M22	90	32.00	20.00	0.47	0.00	0.50	46.80	
131	131	M23	90	32.00	25.00	0.47	0.00	0.50	45.60	
132	132	M24	90	32.00	30.00	0.47	0.00	0.50	45.10	
133	133	M25	90	32.00	0.00	0.44	0.50	0.50	50.50	
134	134	M26	90	32.00	10.00	0.44	0.50	0.50	51.20	
135	135	M27	90	32.00	15.00	0.44	0.50	0.50	51.40	
136	136	M28	90	32.00	20.00	0.44	0.50	0.50	50.60	
137	137	M29	90	32.00	25.00	0.44	0.50	0.50	50.50	
138	138	M30	90	32.00	30.00	0.44	0.50	0.50	47.10	
139	139	M31	90	32.00	0.00	0.40	1.30	0.50	51.40	
140	140	M32	90	32.00	10.00	0.40	1.30	0.50	54.20	
141	141	M33	90	32.00	15.00	0.40	1.30	0.50	54.90	
142	142	M34	90	32.00	20.00	0.40	1.30	0.50	53.80	
143	143	M35	90	32.00	25.00	0.40	1.30	0.50	53.50	
144	144	M36	90	32.00	30.00	0.40	1.30	0.50	50.80	
145	1	N	3	42.50	0.00	0.50	0.00	0.33	34.50	Courard et al. 2003 [81]
146	2	5MK	3	42.50	5.00	0.50	0.00	0.33	34.40	
147	3	10MK	3	42.50	10.00	0.50	0.00	0.33	32.40	
148	4	15MK	3	42.50	15.00	0.50	0.00	0.33	31.70	
149	5	20MK	3	42.50	20.00	0.50	0.00	0.33	27.40	
150	6	N	7	42.50	0.00	0.50	0.00	0.33	47.10	
151	7	5MK	7	42.50	5.00	0.50	0.00	0.33	46.30	
152	8	10MK	7	42.50	10.00	0.50	0.00	0.33	48.60	
153	9	15MK	7	42.50	15.00	0.50	0.00	0.33	47.90	
154	10	20MK	7	42.50	20.00	0.50	0.00	0.33	49.40	
155	11	N	28	42.50	0.00	0.50	0.00	0.33	57.00	
156	12	5MK	28	42.50	5.00	0.50	0.00	0.33	65.10	
157	13	10MK	28	42.50	10.00	0.50	0.00	0.33	70.20	
158	14	15MK	28	42.50	15.00	0.50	0.00	0.33	71.20	
159	15	20MK	28	42.50	20.00	0.50	0.00	0.33	68.40	
160	1	Control	3	53.00	0.00	0.46	0.00	0.50	24.93	Sumasree and Sajja 2016 [82]
161	2	Control	7	53.00	0.00	0.46	0.00	0.50	25.21	
162	3	Control	14	53.00	0.00	0.46	0.00	0.50	25.98	
163	4	Control	28	53.00	0.00	0.46	0.00	0.50	26.39	
164	5	Control	56	53.00	0.00	0.46	0.00	0.50	26.97	
165	6	5MK	3	53.00	5.00	0.46	0.00	0.50	27.21	
166	7	5MK	7	53.00	5.00	0.46	0.00	0.50	28.01	
167	8	5MK	14	53.00	5.00	0.46	0.00	0.50	28.87	
168	9	5MK	28	53.00	5.00	0.46	0.00	0.50	29.36	
169	10	5MK	56	53.00	5.00	0.46	0.00	0.50	30.11	
170	11	10MK	3	53.00	10.00	0.46	0.00	0.50	28.67	
171	12	10MK	7	53.00	10.00	0.46	0.00	0.50	29.12	
172	13	10MK	14	53.00	10.00	0.46	0.00	0.50	29.97	

Table 6 (continued)

No		Sample Code	Input Parameters						Output Parameter	References
Per database	Per work		AS	CG	MK/B	W/B	SP	B/S		
-	-	-	(days)	MPa	(%w/w)	(w/w)	(%w/w)	(w/w)	CS	-
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	MPa	(11)
173	14	10MK	28	53.00	10.00	0.46	0.00	0.50	30.26	
174	15	10MK	56	53.00	10.00	0.46	0.00	0.50	30.99	
175	16	15MK	3	53.00	15.00	0.46	0.00	0.50	29.93	
176	17	15MK	7	53.00	15.00	0.46	0.00	0.50	30.36	
177	18	15MK	14	53.00	15.00	0.46	0.00	0.50	31.01	
178	19	15MK	28	53.00	15.00	0.46	0.00	0.50	31.90	
179	20	15MK	56	53.00	15.00	0.46	0.00	0.50	32.37	
180	21	20MK	3	53.00	20.00	0.46	0.00	0.50	31.19	
181	22	20MK	7	53.00	20.00	0.46	0.00	0.50	31.93	
182	23	20MK	14	53.00	20.00	0.46	0.00	0.50	32.42	
183	24	20MK	28	53.00	20.00	0.46	0.00	0.50	33.11	
184	25	20MK	56	53.00	20.00	0.46	0.00	0.50	33.99	
185	26	25MK	3	53.00	25.00	0.46	0.00	0.50	32.76	
186	27	25MK	7	53.00	25.00	0.46	0.00	0.50	33.51	
187	28	25MK	14	53.00	25.00	0.46	0.00	0.50	33.97	
188	29	25MK	28	53.00	25.00	0.46	0.00	0.50	34.24	
189	30	25MK	56	53.00	25.00	0.46	0.00	0.50	35.71	
190	31	30MK	3	53.00	30.00	0.46	0.00	0.50	30.41	
191	32	30MK	7	53.00	30.00	0.46	0.00	0.50	31.36	
192	33	30MK	14	53.00	30.00	0.46	0.00	0.50	32.99	
193	34	30MK	28	53.00	30.00	0.46	0.00	0.50	33.01	
194	35	30MK	56	53.00	30.00	0.46	0.00	0.50	33.76	
195	1	PC	1	52.50	0.00	0.60	0.00	0.33	17.60	Batis et al. 2005 [2]
196	2	PC	2	52.50	0.00	0.60	0.00	0.33	29.40	
197	3	PC	7	52.50	0.00	0.60	0.00	0.33	44.50	
198	4	PC	28	52.50	0.00	0.60	0.00	0.33	57.50	
199	5	MKC-20	1	52.50	20.00	0.60	0.00	0.33	18.80	
200	6	MKC-20	2	52.50	20.00	0.60	0.00	0.33	29.10	
201	7	MKC-20	7	52.50	20.00	0.60	0.00	0.33	50.40	
202	8	MKC-20	28	52.50	20.00	0.60	0.00	0.33	69.70	
203	9	MK-10	1	52.50	10.00	0.60	0.00	0.33	20.00	
204	10	MK-10	2	52.50	10.00	0.60	0.00	0.33	32.00	
205	11	MK-10	7	52.50	10.00	0.60	0.00	0.33	50.70	
206	12	MK-10	28	52.50	10.00	0.60	0.00	0.33	68.10	
207	13	MK-20	1	52.50	20.00	0.60	0.00	0.33	18.00	
208	14	MK-20	2	52.50	20.00	0.60	0.00	0.33	28.50	
209	15	MK-20	7	52.50	20.00	0.60	0.00	0.33	50.00	
210	16	MK-20	28	52.50	20.00	0.60	0.00	0.33	65.30	
211	1	OPC	1	52.50	0.00	0.36	1.40	0.50	40.00	Kadri et al. 2011 [6]
212	2	OPC	7	52.50	0.00	0.36	1.40	0.50	70.42	
213	3	OPC	28	52.50	0.00	0.36	1.40	0.50	80.42	
214	4	OPC	56	52.50	0.00	0.36	1.40	0.50	84.17	
215	5	MK1	1	52.50	10.00	0.36	2.35	0.50	35.21	

Table 6 (continued)

No		Sample Code	Input Parameters						Output Parameter CS	References
Per database	Per work		AS	CG	MK/B	W/B	SP	B/S		
-	-	-	(days)	MPa	(%w/w)	(w/w)	(%w/w)	(w/w)	MPa	-
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
216	6	MK1	7	52.50	10.00	0.36	2.35	0.50	74.58	
217	7	MK1	28	52.50	10.00	0.36	2.35	0.50	95.00	
218	8	MK1	56	52.50	10.00	0.36	2.35	0.50	96.25	
219	9	MK2	1	52.50	10.00	0.36	2.04	0.50	35.63	
220	10	MK2	7	52.50	10.00	0.36	2.04	0.50	82.71	
221	11	MK2	28	52.50	10.00	0.36	2.04	0.50	97.29	
222	12	MK2	56	52.50	10.00	0.36	2.04	0.50	99.17	
223	1	Control	7	42.50	0.00	0.49	0.00	0.36	37.27	Mardani-Aghabaglou et al. 2014 [83]
224	2	Control	28	42.50	0.00	0.49	0.00	0.36	50.05	
225	3	Control	90	42.50	0.00	0.49	0.00	0.36	56.76	
226	4	MK10%	7	42.50	10.00	0.49	0.00	0.36	38.22	
227	5	MK10%	28	42.50	10.00	0.49	0.00	0.36	48.90	
228	6	MK10%	90	42.50	10.00	0.49	0.00	0.36	54.76	
229	1	0%	1	52.50	0.00	0.38	0.00	0.33	23.40	Potgieter-Vermaak and Potgieter 2006 [84]
230	2	0%	2	52.50	0.00	0.38	0.00	0.33	39.40	
231	3	0%	7	52.50	0.00	0.38	0.00	0.33	47.40	
232	4	0%	14	52.50	0.00	0.38	0.00	0.33	48.80	
233	5	0%	28	52.50	0.00	0.38	0.00	0.33	50.90	
234	6	10%	1	52.50	10.00	0.38	0.00	0.33	51.30	
235	7	10%	2	52.50	10.00	0.38	0.00	0.33	60.60	
236	8	10%	7	52.50	10.00	0.38	0.00	0.33	73.10	
237	9	10%	14	52.50	10.00	0.38	0.00	0.33	85.90	
238	10	10%	28	52.50	10.00	0.38	0.00	0.33	92.80	
239	11	20%	1	52.50	20.00	0.38	0.00	0.33	58.10	
240	12	20%	2	52.50	20.00	0.38	0.00	0.33	60.60	
241	13	20%	7	52.50	20.00	0.38	0.00	0.33	78.80	
242	14	20%	14	52.50	20.00	0.38	0.00	0.33	89.10	
243	15	20%	28	52.50	20.00	0.38	0.00	0.33	92.50	
244	16	30%	1	52.50	30.00	0.38	0.00	0.33	59.10	
245	17	30%	2	52.50	30.00	0.38	0.00	0.33	62.20	
246	18	30%	7	52.50	30.00	0.38	0.00	0.33	80.30	
247	19	30%	14	52.50	30.00	0.38	0.00	0.33	85.00	
248	20	30%	28	52.50	30.00	0.38	0.00	0.33	88.40	
249	1	Cref	3	52.50	0.00	0.45	0.00	0.33	47.94	Cizer et al. 2008 [85]
250	2	Cref	7	52.50	0.00	0.45	0.00	0.33	57.20	
251	3	Cref	28	52.50	0.00	0.45	0.00	0.33	66.23	
252	4	Cref	60	52.50	0.00	0.45	0.00	0.33	64.60	
253	5	Cref	90	52.50	0.00	0.45	0.00	0.33	66.01	
254	1	REF	7	32.00	0.00	0.48	0.00	0.36	6.06	Al-Chaar et al. 2013 [86]
255	2	REF	28	32.00	0.00	0.48	0.00	0.36	10.49	
256	3	REF	90	32.00	0.00	0.48	0.00	0.36	10.73	

Table 6 (continued)

No		Sample Code	Input Parameters						Output Parameter	References
Per database	Per work		AS	CG	MK/B	W/B	SP	B/S		
-	-	-	(days)	MPa	(%w/w)	(w/w)	(%w/w)	(w/w)	CS MPa	-
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
257	1	M0	14	42.50	0.00	0.55	0.00	0.40	40.08	Khatib and Wild 1998 [8]
258	2	M5	14	42.50	5.00	0.55	0.00	0.40	44.79	
259	3	M10	14	42.50	10.00	0.55	0.00	0.40	56.76	
260	4	M15	14	42.50	15.00	0.55	0.00	0.40	56.58	Curcio et al. 1998 [87]
261	1	C	7	52.50	0.00	0.33	0.03	0.51	85.99	
262	2	C	28	52.50	0.00	0.33	0.03	0.51	94.30	
263	3	C	90	52.50	0.00	0.33	0.03	0.51	106.28	
264	4	M1	7	52.50	15.15	0.33	0.03	0.51	100.15	
265	5	M1	28	52.50	15.15	0.33	0.03	0.51	111.29	
266	6	M1	90	52.50	15.15	0.33	0.03	0.51	112.99	
267	7	M2	7	52.50	15.15	0.33	0.03	0.51	101.66	
268	8	M2	28	52.50	15.15	0.33	0.03	0.51	109.70	
269	9	M2	90	52.50	15.15	0.33	0.03	0.51	112.87	
270	10	M3	7	52.50	15.15	0.33	0.03	0.51	97.38	Cyr et al. 2007 [88]
271	11	M3	28	52.50	15.15	0.33	0.03	0.51	111.17	
272	12	M3	90	52.50	15.15	0.33	0.03	0.51	115.25	
273	13	M4	7	52.50	15.15	0.33	0.03	0.51	88.61	
274	14	M4	28	52.50	15.15	0.33	0.03	0.51	103.27	
275	15	M4	90	52.50	15.15	0.33	0.03	0.51	111.72	
276	1	Mref	1	53.50	0.00	0.50	0.00	0.33	24.80	
277	2	Mref	7	53.50	0.00	0.50	0.00	0.33	49.50	
278	3	Mref	28	53.50	0.00	0.50	0.00	0.33	59.80	
279	4	Mref	90	53.50	0.00	0.50	0.00	0.33	62.80	
280	5	Mref-mk	1	53.50	25.00	0.50	0.00	0.33	18.10	Saidat et al. 2012 [89]
281	6	Mref-mk	7	53.50	25.00	0.50	0.00	0.33	45.60	
282	7	Mref-mk	28	53.50	25.00	0.50	0.00	0.33	62.80	
283	8	Mref-mk	90	53.50	25.00	0.50	0.00	0.33	67.50	
284	1	Ai	0.67	52.50	0.00	0.50	0.00	0.33	16.00	
285	2	Ai	1	52.50	0.00	0.50	0.00	0.33	23.40	
286	3	Ai	2	52.50	0.00	0.50	0.00	0.33	35.80	
287	4	Ai	7	52.50	0.00	0.50	0.00	0.33	52.20	
288	5	Ai	28	52.50	0.00	0.50	0.00	0.33	63.00	
289	6	Ai-25MK	0.67	52.50	25.00	0.50	0.00	0.33	8.60	
290	7	Ai-25MK	1	52.50	25.00	0.50	0.00	0.33	15.20	
291	8	Ai-25MK	2	52.50	25.00	0.50	0.00	0.33	24.00	
292	9	Ai-25MK	7	52.50	25.00	0.50	0.00	0.33	47.20	
293	10	Ai-25MK	28	52.50	25.00	0.50	0.00	0.33	64.50	
294	11	Aii	0.67	52.50	0.00	0.50	0.00	0.33	10.80	
295	12	Aii	1	52.50	0.00	0.50	0.00	0.33	15.10	
296	13	Aii	2	52.50	0.00	0.50	0.00	0.33	32.80	
297	14	Aii	7	52.50	0.00	0.50	0.00	0.33	46.10	
298	15	Aii	28	52.50	0.00	0.50	0.00	0.33	55.70	
299	16	Aii-25MK	0.67	52.50	25.00	0.50	0.00	0.33	6.30	
300	17	Aii-25MK	1	52.50	25.00	0.50	0.00	0.33	8.30	
301	18	Aii-25MK	2	52.50	25.00	0.50	0.00	0.33	19.80	

Table 6 (continued)

No		Sample Code	Input Parameters						Output Parameter	References
Per database	Per work		AS	CG	MK/B	W/B	SP	B/S		
-	-	-	(days)	MPa	(%w/w)	(w/w)	(%w/w)	(w/w)	CS	-
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	MPa	(11)
302	19	Aii-25MK	7	52.50	25.00	0.50	0.00	0.33	39.60	
303	20	Aii-25MK	28	52.50	25.00	0.50	0.00	0.33	61.90	
304	21	Bi	0.67	52.50	0.00	0.50	0.00	0.33	17.70	
305	22	Bi	1	52.50	0.00	0.50	0.00	0.33	27.60	
306	23	Bi	2	52.50	0.00	0.50	0.00	0.33	42.80	
307	24	Bi	7	52.50	0.00	0.50	0.00	0.33	54.50	
308	25	Bi	28	52.50	0.00	0.50	0.00	0.33	63.60	
309	26	Bi-25MK	0.67	52.50	25.00	0.50	0.00	0.33	11.90	
310	27	Bi-25MK	1	52.50	25.00	0.50	0.00	0.33	16.70	
311	28	Bi-25MK	2	52.50	25.00	0.50	0.00	0.33	28.70	
312	29	Bi-25MK	7	52.50	25.00	0.50	0.00	0.33	45.40	
313	30	Bi-25MK	28	52.50	25.00	0.50	0.00	0.33	55.90	
314	31	Bii	0.67	42.50	0.00	0.50	0.00	0.33	5.20	
315	32	Bii	1	42.50	0.00	0.50	0.00	0.33	17.10	
316	33	Bii	2	42.50	0.00	0.50	0.00	0.33	28.30	
317	34	Bii	7	42.50	0.00	0.50	0.00	0.33	44.40	
318	35	Bii	28	42.50	0.00	0.50	0.00	0.33	52.00	
319	36	Bii-25MK	0.67	42.50	25.00	0.50	0.00	0.33	4.10	
320	37	Bii-25MK	1	42.50	25.00	0.50	0.00	0.33	10.00	
321	38	Bii-25MK	2	42.50	25.00	0.50	0.00	0.33	20.00	
322	39	Bii-25MK	7	42.50	25.00	0.50	0.00	0.33	43.00	
323	40	Bii-25MK	28	42.50	25.00	0.50	0.00	0.33	51.60	
324	1	Control 7	7	43.00	0.00	0.40	0.00	0.33	26.37	Ashok et al. 2017 [90]
325	2	Control 28	28	43.00	0.00	0.40	0.00	0.33	43.13	
326	3	Control 56	56	43.00	0.00	0.40	0.00	0.33	46.70	
327	4	Control 90	90	43.00	0.00	0.40	0.00	0.33	48.08	
328	1	MK0 1	1	32.50	0.00	0.45	0.00	0.33	11.75	Geng and Li 2017 [91]
329	2	MK3 1	1	32.50	3.00	0.45	0.00	0.33	9.61	
330	3	MK5 1	1	32.50	5.00	0.45	0.00	0.33	6.01	
331	4	MK7 1	1	32.50	7.00	0.45	0.00	0.33	17.49	
332	5	MK10 1	1	32.50	10.00	0.45	0.00	0.33	9.74	
333	6	MK0 3	3	32.50	0.00	0.45	0.00	0.33	38.72	
334	7	MK3 3	3	32.50	3.00	0.45	0.00	0.33	41.52	
335	8	MK5 3	3	32.50	5.00	0.45	0.00	0.33	40.99	
336	9	MK7 3	3	32.50	7.00	0.45	0.00	0.33	38.32	
337	10	MK10 3	3	32.50	10.00	0.45	0.00	0.33	44.33	
338	11	MK0 7	7	32.50	0.00	0.45	0.00	0.33	48.31	
339	12	MK3 7	7	32.50	3.00	0.45	0.00	0.33	57.80	
340	13	MK5 7	7	32.50	5.00	0.45	0.00	0.33	56.19	
341	14	MK7 7	7	32.50	7.00	0.45	0.00	0.33	64.74	
342	15	MK10 7	7	32.50	10.00	0.45	0.00	0.33	67.68	
343	16	MK0 28	28	32.50	0.00	0.45	0.00	0.33	68.65	
344	17	MK3 28	28	32.50	3.00	0.45	0.00	0.33	77.73	
345	18	MK5 28	28	32.50	5.00	0.45	0.00	0.33	74.40	
346	19	MK7 28	28	32.50	7.00	0.45	0.00	0.33	77.74	
347	20	MK10 28	28	32.50	10.00	0.45	0.00	0.33	78.54	

Table 6 (continued)

No		Sample Code	Input Parameters						Output Parameter	References
Per database	Per work		AS	CG	MK/B	W/B	SP	B/S		
-	-	-	(days)	MPa	(%w/w)	(w/w)	(%w/w)	(w/w)	CS	-
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	MPa	(11)
(10)										
348	1	RM I	28	42.50	0.00	0.33	1.00	0.33	76.40	Pavlikova et al. 2009 [5]
349	2	CM16 A	28	42.50	5.00	0.33	1.00	0.33	60.47	
350	3	CM16 B	28	42.50	5.00	0.33	2.00	0.33	74.99	
351	4	CM17	28	42.50	10.00	0.33	2.00	0.33	74.08	
352	5	CM18	28	42.50	15.00	0.33	2.00	0.33	75.98	
353	6	RMII	28	42.50	0.00	0.45	0.00	0.33	55.64	
354	7	CM7	28	42.50	5.00	0.45	0.00	0.33	66.95	
355	8	CM8	28	42.50	10.00	0.45	0.00	0.33	70.89	
356	9	CM9	28	42.50	15.00	0.45	0.00	0.33	62.53	
357	10	RM III	28	42.50	0.00	0.55	0.00	0.33	44.98	
358	11	CM10	28	42.50	5.00	0.55	0.00	0.33	48.46	
359	12	CM11	28	42.50	10.00	0.55	0.00	0.33	52.29	
360	13	CM12	28	42.50	15.00	0.55	0.00	0.33	53.47	
361	14	RM IV	28	52.50	0.00	0.33	1.00	0.33	76.97	
362	15	CM13	28	52.50	5.00	0.33	1.00	0.33	77.61	
363	16	CM14 A	28	52.50	10.00	0.33	1.00	0.33	82.96	
364	17	CM15 A	28	52.50	15.00	0.33	1.00	0.33	83.16	
365	18	CM14 B	28	52.50	10.00	0.33	2.00	0.33	81.11	
366	19	CM15 B	28	52.50	15.00	0.33	2.00	0.33	77.56	
367	20	RM V	28	52.50	0.00	0.45	0.00	0.33	64.05	
368	21	CM1	28	52.50	5.00	0.45	0.00	0.33	64.20	
369	22	CM2	28	52.50	10.00	0.45	0.00	0.33	69.26	
370	23	CM3	28	52.50	15.00	0.45	0.00	0.33	71.51	
371	24	RM VI	28	52.50	0.00	0.55	0.00	0.33	52.11	
372	25	CM4	28	52.50	5.00	0.55	0.00	0.33	49.42	
373	26	CM5	28	52.50	10.00	0.55	0.00	0.33	59.14	
374	27	CM6	28	52.50	15.00	0.55	0.00	0.33	61.43	
375	1	0	28	42.50	0.00	0.60	0.00	0.33	10.43	Khater 2010 [92]
376	2	10	28	42.50	10.00	0.60	0.00	0.33	11.22	
377	3	15	28	42.50	15.00	0.60	0.00	0.33	11.44	
378	4	20	28	42.50	20.00	0.60	0.00	0.33	11.66	
379	5	25	28	42.50	25.00	0.60	0.00	0.33	11.78	
380	6	30	28	42.50	30.00	0.60	0.00	0.33	9.75	Parande et al. 2008 [93]
381	1	N	3	42.50	0.00	0.40	0.00	0.33	22.00	
382	2	C + 5MK	3	42.50	5.00	0.40	0.00	0.33	31.50	
383	3	C + 10MK	3	42.50	10.00	0.40	0.00	0.33	30.00	
384	4	C + 15MK	3	42.50	15.00	0.40	0.00	0.33	29.00	
385	5	C + 20MK	3	42.50	20.00	0.40	0.00	0.33	27.00	
386	6	N	7	42.50	0.00	0.40	0.00	0.33	32.50	
387	7	C + 5MK	7	42.50	5.00	0.40	0.00	0.33	40.00	
388	8	C + 10MK	7	42.50	10.00	0.40	0.00	0.33	43.50	
389	9	C + 15MK	7	42.50	15.00	0.40	0.00	0.33	42.10	
390	10	C + 20MK	7	42.50	20.00	0.40	0.00	0.33	44.00	
391	11	N	28	42.50	0.00	0.40	0.00	0.33	40.00	
392	12	C + 5MK	28	42.50	5.00	0.40	0.00	0.33	52.00	

Table 6 (continued)

No		Sample Code	Input Parameters						Output Parameter	References
Per database	Per work		AS	CG	MK/B	W/B	SP	B/S		
-	-	-	(days)	MPa	(%w/w)	(w/w)	(%w/w)	(w/w)	CS	-
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	MPa	(11)
393	13	C + 10MK	28	42.50	10.00	0.40	0.00	0.33	56.00	
394	14	C + 15MK	28	42.50	15.00	0.40	0.00	0.33	60.00	
395	15	C + 20MK	28	42.50	20.00	0.40	0.00	0.33	58.00	
396	16	N	90	42.50	0.00	0.40	0.00	0.33	48.00	
397	17	C + 5MK	90	42.50	5.00	0.40	0.00	0.33	52.00	
398	18	C + 10MK	90	42.50	10.00	0.40	0.00	0.33	64.00	
399	19	C + 15MK	90	42.50	15.00	0.40	0.00	0.33	67.00	
400	20	C + 20MK	90	42.50	20.00	0.40	0.00	0.33	65.00	
401	1	MR1	1	52.50	0.00	0.30	0.01	0.33	31.10	Mansour et al. 2012 [94]
402	2	MR3	3	52.50	0.00	0.30	0.01	0.33	77.24	
403	3	MR7	7	52.50	0.00	0.30	0.01	0.33	82.45	
404	4	MR14	14	52.50	0.00	0.30	0.01	0.33	92.50	
405	5	MR21	21	52.50	0.00	0.30	0.01	0.33	79.92	
406	6	MR28	28	52.50	0.00	0.30	0.01	0.33	83.06	
407	7	MMK1	1	52.50	10.00	0.30	0.01	0.33	22.99	
408	8	MMK3	3	52.50	10.00	0.30	0.01	0.33	66.37	
409	9	MMK7	7	52.50	10.00	0.30	0.01	0.33	75.90	
410	10	MMK14	14	52.50	10.00	0.30	0.01	0.33	85.25	
411	11	MMK21	21	52.50	10.00	0.30	0.01	0.33	71.98	
412	12	MMK28	28	52.50	10.00	0.30	0.01	0.33	78.75	
413	1	MK0	7	42.50	0.00	0.45	0.50	0.50	37.00	Lee et al. 2005 [95]
414	2	MK0	28	42.50	0.00	0.45	0.50	0.50	48.60	
415	3	MK0	91	42.50	0.00	0.45	0.50	0.50	56.80	
416	4	MK5	7	42.50	5.00	0.45	0.50	0.50	33.80	
417	5	MK5	28	42.50	5.00	0.45	0.50	0.50	46.30	
418	6	MK5	91	42.50	5.00	0.45	0.50	0.50	55.50	
419	7	MK10	7	42.50	10.00	0.45	0.40	0.50	43.20	
420	8	MK10	28	42.50	10.00	0.45	0.40	0.50	50.40	
421	9	MK10	91	42.50	10.00	0.45	0.40	0.50	56.90	
422	10	MK15	7	42.50	15.00	0.45	0.40	0.50	41.80	
423	11	MK15	28	42.50	15.00	0.45	0.40	0.50	51.20	
424	12	MK15	91	42.50	15.00	0.45	0.40	0.50	63.00	

AS, age of specimen; CG, cement grade; MK/B, metakaolin percentage in relation to total binder, W/B, water-to-binder ratio; SP, superplasticizer; B/S, binder-to-sand ratio; CS, cement-based mortar compressive strength

Declarations

Conflict of interest The authors confirm that this article content has no conflict of interest.

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